

Bodies, Configurations, and Motions

Kinematics of a deforming body: motion, the material
derivative,
the deformation gradient, and mass

A from-scratch, definition–theorem–proof companion to
Steigmann and Shirani, *Principles of Continuum Mechanics*,
Chapter 2
(§ 2.1 through § 2.5, in the book’s order and numbering)

Every figure of the text is described in an intuitive remark, every equation is derived using only the algebra of Chapter 1, and the worked example is solved in full. Section and equation numbers match the book.

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Chapter 2

Bodies, Configurations, and Motions

This chapter builds the *kinematics* of continuum mechanics: the language for describing where the material of a body is, how fast it moves, and how it deforms, all before any mention of force. The single new idea is that we carry *two* pictures of the body at once, a fixed reference copy and the moving current one, and translate between them. Everything computational rests on the vector and tensor algebra of Chapter 1; we cite it freely and prove each new relation from it.

2.1 Position, velocity and acceleration

A *body* B is a collection of *material points*, labelled p . In the older literature these are sometimes called *particles*, to convey the notion of a small indivisible object; that name has some heuristic value but is not really what is intended. A material point is better regarded as a *primitive* notion: not a tiny lump of matter, but an abstract label for “a bit of the material”. The body B itself is not a region of space; we never see B directly, only the region of space it fills at a given instant. In the book’s figures B is drawn with a *dashed* outline for exactly this reason (Figure 2.1): it is a set of labels, not a place.

REMARK 2.1 (Figures 2.1–2.3: body versus configuration). Figure 2.1

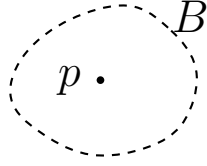


Figure 2.1: A material body.

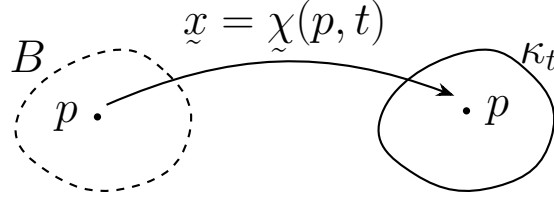


Figure 2.2: Configuration of the body at time t .

shows the abstract body B (a dashed blob with a marked point p): the reservoir of material points. Figures 2.2 and 2.3 each show B on the left mapped by an arrow to a *solid* blob on the right, the difference being only which placement of the body the arrow produces: the *current* configuration κ_t at time t in Figure 2.2, and a fixed *reference* configuration κ in Figure 2.3. Solid outlines mark genuine regions of Euclidean space; the dashed B is not one. The whole chapter is the art of never confusing the label p with either of its placements $\underline{x}$ or $\underline{X}$.

DEFINITION 2.2 (Configuration and motion). At time t the body occupies a *configuration* κ_t , a region of the Euclidean point space $\mathcal{E}^3$, and $p \in B$ sits at the position $\underline{x} \in \kappa_t$. Distinct material points occupy distinct positions at a fixed time, so there is an invertible *motion* $\underline{\chi}(p, t)$ with

$$\underline{x} = \underline{\chi}(p, t), \quad p = \underline{\chi}^{-1}(\underline{x}, t). \quad (2.1)$$

At fixed t , $\underline{\chi}$ tells us where each material point is, and $\underline{\chi}^{-1}$ tells us

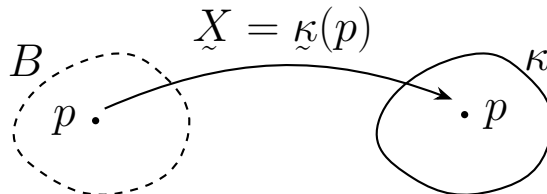


Figure 2.3: A reference configuration for the body.

which material point sits at a given place. Freezing p and letting t run traces the path of one material point through space; its time derivatives are its velocity and acceleration.

DEFINITION 2.3 (Velocity, acceleration).

$$\underline{v}(p, t) = \frac{\partial}{\partial t} \underline{\chi}(p, t), \quad \underline{a}(p, t) = \frac{\partial^2}{\partial t^2} \underline{\chi}(p, t). \quad (2.2)$$

The derivatives are taken at fixed p : they follow the *same* material point.

Working with the ethereal B is awkward, because a “label” has no coordinates. The remedy is to name each material point by the fixed place it occupies in one chosen, frozen configuration, the *reference* configuration κ .

DEFINITION 2.4 (Reference configuration). Fix a configuration κ and let $\underline{X} \in \kappa$ be positions in one-to-one correspondence with material points through an invertible placement map $\underline{\kappa}$,

$$\underline{X} = \underline{\kappa}(p), \quad p = \underline{\kappa}^{-1}(\underline{X}). \quad (2.3)$$

A common, though not obligatory, choice is the configuration actually occupied at some instant t_0 , $\kappa = \kappa_{t_0}$, for which

$$\underline{\kappa}(p) = \underline{\chi}(p, t_0). \quad (2.4)$$

Composing the placement’s inverse with the motion expresses the current position directly in terms of the reference position (Figure 2.4):

$$\underline{x} = \underline{\chi}_\kappa(\underline{X}, t), \quad \underline{\chi}_\kappa(\underline{X}, t) = \underline{\chi}(\underline{\kappa}^{-1}(\underline{X}), t). \quad (2.5\text{--}2.6)$$

The subscript records the dependence on the chosen κ . Since each p corresponds to exactly one $\underline{x} \in \kappa_t$ and one $\underline{X} \in \kappa$, the map $\underline{\chi}_\kappa(\cdot, t)$ is invertible at each t ,

$$\underline{X} = \underline{\chi}_\kappa^{-1}(\underline{x}, t). \quad (2.7)$$

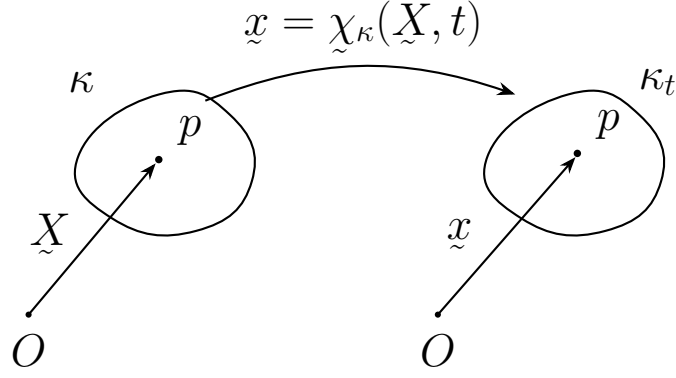


Figure 2.4: Mapping of reference configuration to current configuration.

REMARK 2.5 (Figure 2.4: the deformation map). Figure 2.4 draws two *solid* blobs, κ (reference, left) and κ_t (current, right), with the same material point p appearing in each, at $\tilde{X}$ and at $\tilde{x}$. The connecting arrow is χ_κ : it is the object a deformation actually is, a rule sending each reference place to its current place. Because both blobs are real regions of space, this map is the one we can differentiate, and its gradient will be the deformation gradient of §2.4.

Because B and κ are in fixed one-to-one correspondence, using (p, t) as independent variables is the same as using $(\tilde{X}, t)$. The velocity therefore has a *referential* form (a function of $\tilde{X}, t$) and a *spatial* form (a function of $\tilde{x}, t$),

$$\hat{v}(\tilde{X}, t) = \underset{\sim}{v}(\kappa^{-1}(\tilde{X}), t) = \frac{\partial}{\partial t} \chi_\kappa(\tilde{X}, t), \quad \tilde{v}(\tilde{x}, t) = \underset{\sim}{v}(\chi_\kappa^{-1}(\tilde{x}, t), t), \quad (2.8-2.9)$$

the partial in (2.8) being taken at fixed $\tilde{X}$, i.e. at fixed p . The two forms are related through (2.5) and (2.7),

$$\hat{v}(\tilde{X}, t) = \tilde{v}(\chi_\kappa(\tilde{X}, t), t), \quad \tilde{v}(\tilde{x}, t) = \hat{v}(\chi_\kappa^{-1}(\tilde{x}, t), t). \quad (2.10)$$

2.2 Material, referential and spatial descriptions

Any field f carried by the material, temperature, density, velocity, may be written as a function of the material point, of the current position, or

of the reference position:

$$f = \bar{f}(p, t) = \tilde{f}(\underline{x}, t) = \hat{f}(\underline{X}, t), \quad (2.11)$$

where

$$\tilde{f}(\underline{x}, t) = \bar{f}(\chi^{-1}(\underline{x}, t), t), \quad \hat{f}(\underline{X}, t) = \bar{f}(\kappa^{-1}(\underline{X}), t), \quad (2.12)$$

and, relating the last two directly,

$$\hat{f}(\underline{X}, t) = \tilde{f}(\chi_{\kappa}(\underline{X}, t), t), \quad \tilde{f}(\underline{x}, t) = \hat{f}(\chi_{\kappa}^{-1}(\underline{x}, t), t). \quad (2.13)$$

These are the *material*, *spatial*, and *referential* descriptions of f .

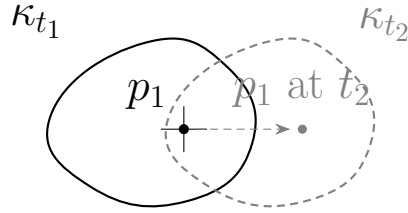
REMARK 2.6 (same value, three different functions: why the decorations matter). This is the single most important notational point of the chapter, and it is the exact analogue of the caret in Chapter 1, where $\hat{\varphi}(x_1, x_2, x_3)$ and $\varphi(\underline{x})$ shared their values but had different domains ($\mathbb{R}^3$ versus a point space) and so were different functions. Here the *same number* f , the temperature of a given bit of matter at a given instant, is produced by *three different machines*:

$$\begin{aligned} \bar{f} &\text{ eats a material label } p, \\ \tilde{f} &\text{ eats a current place } \underline{x}, \\ \hat{f} &\text{ eats a reference place } \underline{X}. \end{aligned}$$

They agree in value only when p , $\underline{x}$, $\underline{X}$ describe the same material point at the same time, i.e. when $\underline{x} = \chi_{\kappa}(\underline{X}, t)$. Their *derivatives* are genuinely different, because “hold the variable fixed” means hold p , hold $\underline{x}$, or hold $\underline{X}$, and these are three different physical instructions. The bar, tilde and caret are the honest bookkeeping of which machine is meant; the book (and we) will soon drop the decorations “to minimize visual distraction”, exactly as the caret was dropped in Chapter 1, on the understanding that the reader remembers which function a symbol denotes.

In the *spatial* description $\underline{x}$ and t are the independent variables. Fixing $\underline{x}$ fixes a *place*, not a material point, so as time passes different

material points stream through that fixed $\underline{x}$ (Figure 2.5). The set of fixed positions $\underline{x}$ is a *control volume*, a window in space through which material flows.



$\underline{x}$ stays put; at t_2 a *different* point p_2 sits there

Figure 2.5: Different material points occupy position $\underline{x}$ at different times.

REMARK 2.7 (Figure 2.5: the control-volume viewpoint). Figure 2.5 shows one fixed position $\underline{x}$ occupied by different material points at different times: the Eulerian, or spatial, standpoint of an observer nailed to space (a thermometer on a wall) rather than riding with the matter (a thermometer floating down a river). The distinction between “rate of change at a fixed place” and “rate of change following the matter” is precisely what the material derivative below measures.

Take the differential of f in each description. Using the coordinate free chain rule of Chapter 1, $df = \text{grad } f \cdot d\underline{x}$ (there written $d\varphi = \text{grad } \varphi \cdot d\underline{x}$), and allowing time to vary,

$$df = \tilde{f}' dt + \text{grad } \tilde{f} \cdot d\underline{x}, \quad (2.14)$$

$$df = (\hat{f})^\bullet dt + \text{Grad } \hat{f} \cdot d\underline{X}. \quad (2.15)$$

Here $\tilde{f}' = \partial \tilde{f} / \partial t$ is the *spatial time derivative* (at fixed $\underline{x}$) and grad is the gradient in $\underline{x}$; while $(\hat{f})^\bullet = \partial \hat{f} / \partial t$ is the *referential* (equivalently *material*) time derivative (at fixed $\underline{X}$, i.e. fixed p) and Grad is the gradient in $\underline{X}$. Since (2.14) and (2.15) are two expressions for the *same* increment df , dropping decorations,

$$\dot{f} dt + \text{Grad } f \cdot d\underline{X} = f' dt + \text{grad } f \cdot d\underline{x}. \quad (2.16)$$

We call $\dot{f}$ the *material derivative*: the rate of change of f following a fixed material point.

THEOREM 2.8 (Material derivative in the spatial description). *For a scalar field, with $\underline{v}$ the (spatial) material velocity,*

$$\dot{f} = f' + \text{grad } f \cdot \underline{v}. \quad (2.17)$$

For a vector field $\underline{u}$,

$$\dot{\underline{u}} = \underline{u}' + (\text{grad } \underline{u})\underline{v}, \quad (2.20)$$

and in particular the material acceleration is

$$\underline{a} = \dot{\underline{v}} = \underline{v}' + \underline{L}\underline{v}, \quad \underline{L} = \text{grad } \underline{v}. \quad (2.21)$$

Proof. Follow one material point: hold p , hence $\underline{X}$, fixed, so $d\underline{X} = \underline{0}$ in (2.16). Moving with the point, $d\underline{x} = d\underline{\chi}_\kappa(\underline{X}, t) = \underline{v} dt$ (the position changes at the velocity). Substituting into (2.16),

$$\dot{f} dt = f' dt + \text{grad } f \cdot \underline{v} dt = (f' + \text{grad } f \cdot \underline{v}) dt,$$

and cancelling dt gives (2.17). For a vector field the identical argument applied componentwise, or directly from the vector chain rule $d\underline{u} = \underline{u}' dt + (\text{grad } \underline{u})d\underline{x}$ of Chapter 1, gives

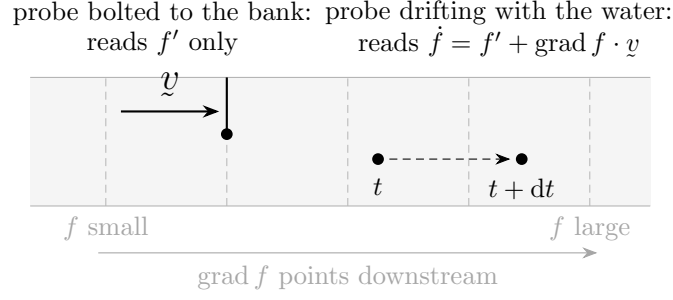
$$\dot{\underline{u}} dt + (\text{Grad } \underline{u})d\underline{X} = d\underline{u} = \underline{u}' dt + (\text{grad } \underline{u})d\underline{x}, \quad (2.18\text{--}2.19)$$

and setting $d\underline{X} = \underline{0}$, $d\underline{x} = \underline{v} dt$ yields (2.20). Taking $\underline{u} = \underline{v}$ gives the acceleration (2.21), with $\underline{L} = \text{grad } \underline{v}$ the spatial velocity gradient. ■

REMARK 2.9 (the convective term, intuitively). The material derivative splits into two clearly physical pieces:

$$\dot{f} = \underbrace{f'}_{\text{local: change at a fixed place}} + \underbrace{\text{grad } f \cdot \underline{v}}_{\text{convective: change from being carried to new places}}.$$

Float down a river whose temperature is steady in time ($f' = 0$) but rises downstream ($\text{grad } f \neq 0$): you still warm up, at the rate $\text{grad } f \cdot \underline{v}$, purely because the current carries you into warmer water. The acceleration (2.21) shows the same effect is why steady flows ($\underline{v}' = 0$) can still accelerate fluid particles, through the nonlinear term $\underline{L}\underline{v} = (\text{grad } \underline{v})\underline{v}$; this convective nonlinearity is the source of most of the difficulty in fluid and solid dynamics.



SUPPLEMENTARY FIGURE A. The two time derivatives of a field. A probe bolted to the bank samples one *place* and records only the explicit time change f' . A probe drifting with the water samples one *material point*; even in a perfectly steady field ($f' = 0$) it registers a change, because the current carries it into regions of different f . That extra reading is the convective term $\text{grad } f \cdot v$, and the sum is the material derivative $\dot{f}$ of (2.17).

2.3 Cartesian coordinates

To compute, we resolve everything on orthonormal bases. The current position uses a fixed basis $\{\underline{e}_i\}$ of the spatial translation space $\mathbb{E}^3$, the reference position a second fixed basis $\{\hat{\underline{E}}_A\}$ of the referential translation space $\hat{\mathbb{E}}^3$ (Figure 2.6):

$$\underline{x} = x_i \underline{e}_i \quad (i \in \{1, 2, 3\}), \quad \underline{X} = X_A \hat{\underline{E}}_A \quad (A \in \{1, 2, 3\}). \quad (2.22-2.23)$$

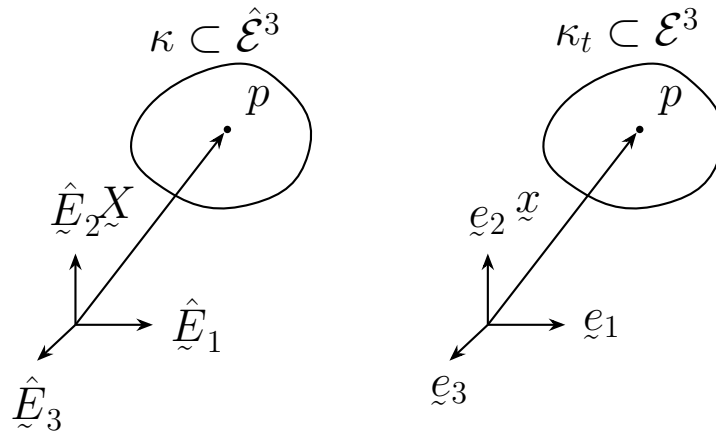


Figure 2.6: Position vectors in κ and κ_t .

REMARK 2.10 (Figure 2.6, and why two bases). Figure 2.6 draws the reference blob $\kappa \subset \hat{\mathcal{E}}^3$ with its own frame $\{\hat{\underline{E}}_A\}$ and the current blob

$\kappa_t \subset \mathcal{E}^3$ with a separate frame $\{\underline{e}_i\}$. Keeping the two frames *conceptually distinct*, even though both are copies of ordinary three-space, is the whole trick of the chapter: a quantity carrying one index of each kind (a lower-case i from $\{\underline{e}_i\}$ and an upper-case A from $\{\hat{\underline{E}}_A\}$) is a *two-point tensor*, an object living with one foot in each configuration. The deformation gradient is the first and most important example. This is why the book uses capital letters X_A , $\hat{\underline{E}}_A$ for reference quantities and lower-case x_i , $\underline{e}_i$ for current ones: the alphabet itself tracks which space you are in.

Regarding χ_κ as a vector in $\mathbb{E}^3$ and using that the $\hat{\underline{E}}_A$ are fixed,

$$x_i \underline{e}_i = \chi_\kappa(X_A \hat{\underline{E}}_A, t) = \chi_i(X_A, t) \underline{e}_i, \quad x_i = \chi_i(X_A, t), \quad (2.24)$$

shorthand for $x_i = \chi_i(X_1, X_2, X_3, t)$. The velocity, in the referential description, is obtained by differencing at fixed $\underline{X}$ and passing to the limit:

$$\begin{aligned} \underline{v} &= \lim_{\alpha \rightarrow 0} \frac{1}{\alpha} [\chi_\kappa(\underline{X}, t + \alpha) - \chi_\kappa(\underline{X}, t)] \\ &= \underline{e}_i \lim_{\alpha \rightarrow 0} \frac{1}{\alpha} [\chi_i(X_A, t + \alpha) - \chi_i(X_A, t)] = \dot{\chi}_i \underline{e}_i, \end{aligned} \quad (2.25-2.26)$$

so $\hat{v}_i = \dot{\chi}_i = \partial \chi_i / \partial t$; likewise $\hat{a}_i = \ddot{\chi}_i = \partial^2 \chi_i / \partial t^2$ (2.27). In the spatial description $\underline{v} = \tilde{v}_i(x_k, t) \underline{e}_i$, $\underline{a} = \tilde{a}_i(x_k, t) \underline{e}_i$ (2.28). Resolving the material derivative (2.21), $\underline{a} = \underline{v}' + (\text{grad } \underline{v}) \underline{v}$, on $\{\underline{e}_i\}$ gives the component identity

$$\tilde{a}_i = \tilde{v}'_i + \tilde{v}_{i,j} \tilde{v}_j, \quad (2.29)$$

and applying (2.20) to the position field $\underline{x}$ itself, with $\underline{x}' = \underline{0}$ and $x_{i,j} = \delta_{ij}$, recovers the sanity check $\dot{x}_i = x_{i,j} \tilde{v}_j = \delta_{ij} \tilde{v}_j = \tilde{v}_i$: the material derivative of position is velocity, as it must be.

REMARK 2.11 (why $\underline{x}' = \underline{0}$, and why this is not circular). This step confuses almost everyone on first reading, because $\underline{x}$ is manifestly moving: the material point does change its position with time. The resolution is that the prime is *not* a material derivative.

Take the field $\underline{u}$ in (2.18) to be *position itself*. Its three descriptions are

$$\underline{\bar{u}}(p, t) = \chi(p, t), \quad \underline{\hat{u}}(\underline{X}, t) = \chi_\kappa(\underline{X}, t), \quad \underline{\tilde{u}}(\underline{x}, t) = \underline{x}.$$

The first two genuinely depend on t : hold the material point fixed and its location changes. The third does not, and the reason is purely definitional. In the spatial description the independent variable *is* $\underline{x}$. Asking for the value of the position field at the place $\underline{x}$ is asking “what position is the position $\underline{x}$?”, and the answer is $\underline{x}$, at every instant. The function $\underline{\tilde{u}}$ is the identity map of $\mathcal{E}^3$, and the identity map carries no t in it at all. Hence

$$\underline{u}' = \left. \frac{\partial}{\partial t} \right|_{\underline{x} \text{ fixed}} \underline{\tilde{u}}(\underline{x}, t) = \left. \frac{\partial \underline{x}}{\partial t} \right|_{\underline{x} \text{ fixed}} = \underline{0}, \quad \text{componentwise } \underline{\tilde{u}}'_i = \frac{\partial x_i}{\partial t} = 0.$$

Nothing is being asserted about the motion here. What is being asserted is that a photograph of space, labelled by its own coordinates, does not change when the clock advances. The matter flowing through the point changes; the point does not.

The second ingredient is $\text{grad } \underline{u} = \underline{I}$, for the same reason: from $\underline{\tilde{u}}_i = x_i$ we get $u_{i,j} = \partial x_i / \partial x_j = \delta_{ij}$, so $\text{grad } \underline{u} = \delta_{ij} \underline{e}_i \otimes \underline{e}_j = \underline{I}$. Substituting both into (2.20),

$$\underline{\dot{u}} = \underline{u}' + (\text{grad } \underline{u})\underline{v} = \underline{0} + \underline{I}\underline{v} = \underline{v},$$

that is $\dot{x}_i = \tilde{v}_i$. Read backwards this is a consistency check on the whole apparatus rather than a new result: (2.17) and (2.20) were constructed precisely so that $\dot{f} = f' + \text{grad } f \cdot \underline{v}$ would reproduce the rate seen by a material point, and the material point’s rate of change of position is, by (2.2)₁, its velocity. Had the formula returned anything other than $\underline{v}$, the definition of the material derivative would have been wrong.

The whole content of the identity is therefore the split

$$\underbrace{\dot{x}_i}_{\text{the point moves}} = \underbrace{x'_i}_{=0: \text{ the place does not move}} + \underbrace{x_{i,j}\tilde{v}_j}_{= \tilde{v}_i: \text{ convection}}.$$

All of the motion sits in the convective term, none of it in the spatial time derivative. This is the cleanest possible illustration of the difference

between $'$ and $\dot{}$: for the position field the two are as far apart as they can be, $\underline{0}$ against $\underline{v}$.

EXAMPLE 2.12 (converting a motion between descriptions). Consider the motion (linear in $\underline{X}$, hence atypical, but transparent)

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} e^t & 0 & 0 \\ 0 & 1 & t \\ 0 & -t & 1 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix}, \quad \text{i.e.} \quad \begin{aligned} x_1 &= e^t X_1, \\ x_2 &= X_2 + tX_3, \\ x_3 &= -tX_2 + X_3. \end{aligned} \quad (2.30)$$

Referential velocity and acceleration. Differentiate at fixed $\underline{X}$:

$$\begin{aligned} \hat{v}_1 &= \dot{x}_1 = e^t X_1, & \hat{v}_2 &= X_3, & \hat{v}_3 &= -X_2; \\ \hat{a}_1 &= e^t X_1, & \hat{a}_2 &= 0, & \hat{a}_3 &= 0. \end{aligned}$$

Inversion. To get the spatial forms we invert (2.30). Immediately $X_1 = e^{-t}x_1$. The lower 2×2 block $\begin{pmatrix} 1 & t \\ -t & 1 \end{pmatrix}$ has determinant $1 + t^2$ and inverse $\frac{1}{1+t^2} \begin{pmatrix} 1 & -t \\ t & 1 \end{pmatrix}$, so

$$X_1 = e^{-t}x_1, \quad X_2 = \frac{x_2 - tx_3}{1 + t^2}, \quad X_3 = \frac{tx_2 + x_3}{1 + t^2}. \quad (2.31)$$

Spatial velocity. Substitute (2.31) into $\hat{v}_i$:

$$\begin{aligned} \tilde{v}_1 &= e^t X_1 = e^t e^{-t} x_1 = x_1, \\ \tilde{v}_2 &= X_3 = \frac{tx_2 + x_3}{1 + t^2}, & \tilde{v}_3 &= -X_2 = \frac{tx_3 - x_2}{1 + t^2}. \end{aligned}$$

Spatial acceleration, directly. Substitute (2.31) into $\hat{a}_i$: $\tilde{a}_1 = e^t X_1 = x_1$, $\tilde{a}_2 = 0$, $\tilde{a}_3 = 0$, i.e. $\underline{a} = x_1 \underline{e}_1$.

Spatial acceleration, via (2.29) (a check on the material derivative). The spatial velocity gradient is

$$(\tilde{v}_{i,j}) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{t}{1+t^2} & \frac{1}{1+t^2} \\ 0 & -\frac{1}{1+t^2} & \frac{t}{1+t^2} \end{pmatrix}, \quad (2.33)$$

and the spatial time derivatives $\tilde{v}'_i = \partial \tilde{v}_i / \partial t$ (at fixed $\underline{x}$) are $\tilde{v}'_1 = 0$ and, by the quotient rule,

$$\tilde{v}'_2 = \frac{(1 - t^2)x_2 - 2tx_3}{(1 + t^2)^2}, \quad \tilde{v}'_3 = \frac{(1 - t^2)x_3 + 2tx_2}{(1 + t^2)^2}. \quad (2.32)$$

Then $\tilde{a}_2 = \tilde{v}'_2 + \tilde{v}_{2,j}\tilde{v}_j = \tilde{v}'_2 + \tilde{v}_{2,2}\tilde{v}_2 + \tilde{v}_{2,3}\tilde{v}_3$; with $\tilde{v}_{2,2} = t/(1 + t^2)$, $\tilde{v}_{2,3} = 1/(1 + t^2)$,

$$\tilde{v}_{2,2}\tilde{v}_2 + \tilde{v}_{2,3}\tilde{v}_3 = \frac{t(tx_2 + x_3) + (tx_3 - x_2)}{(1 + t^2)^2} = \frac{(t^2 - 1)x_2 + 2tx_3}{(1 + t^2)^2} = -\tilde{v}'_2,$$

so $\tilde{a}_2 = 0$, matching the direct computation; likewise $\tilde{a}_1 = x_1$, $\tilde{a}_3 = 0$. The two routes agree, which is the point of the example.

REMARK 2.13 (a correction to the printed $\tilde{v}'_2$). The material derivative check forces $\tilde{v}'_2 = [(1 - t^2)x_2 - 2tx_3]/(1 + t^2)^2$: only this value makes $\tilde{a}_2 = \tilde{v}'_2 + \tilde{v}_{2,j}\tilde{v}_j$ vanish, as it must since $\hat{a}_2 = 0$. (The text's printed coefficient of x_2 in $\tilde{v}'_2$ is a misprint; the denominator is squared and the numerator is $1 - t^2$, not $1 - 2t^2$.)

2.4 The deformation gradient

Assume the motion $\chi_\kappa(\underline{X}, t)$ is differentiable in $\underline{X}$. Holding t fixed, its linear part is a tensor, the *deformation gradient*:

$$d\underline{x} = \underline{F} d\underline{X}, \quad \underline{F} = \text{Grad } \chi_\kappa. \quad (2.34)$$

Allowing t to vary as well,

$$d\underline{x} = \underline{F} d\underline{X} + \underline{v} dt. \quad (2.35)$$

THEOREM 2.14 (The two gradients of a field are related by $\underline{F}$). *For any field u ,*

$$\text{Grad } u = (\text{grad } u)\underline{F}. \quad (2.37)$$

Proof. Write the increment $d\mathbf{u}$ referentially and spatially, as in (2.19), and substitute (2.35) and the material derivative $\dot{\mathbf{u}} = \mathbf{u}' + (\text{grad } \mathbf{u})\mathbf{v}$:

$$\dot{\mathbf{u}} dt + (\text{Grad } \mathbf{u})d\mathbf{X} = [\mathbf{u}' + (\text{grad } \mathbf{u})\mathbf{v}] dt + (\text{grad } \mathbf{u})\mathbf{F} d\mathbf{X}. \quad (2.36)$$

The dt terms cancel by (2.20), leaving $(\text{Grad } \mathbf{u})d\mathbf{X} = (\text{grad } \mathbf{u})\mathbf{F} d\mathbf{X}$ for every $d\mathbf{X}$; hence $\text{Grad } \mathbf{u} = (\text{grad } \mathbf{u})\mathbf{F}$. $\blacksquare$

To read off components, differentiate $\mathbf{x} = \chi_i \mathbf{e}_i$ at fixed t using $d\chi_i = \chi_{i,A} dX_A = \chi_{i,A} \hat{\mathbf{E}}_A \cdot d\mathbf{X}$:

$$d\mathbf{x} = \chi_{i,A} \mathbf{e}_i \hat{\mathbf{E}}_A \cdot d\mathbf{X} = (\chi_{i,A} \mathbf{e}_i \otimes \hat{\mathbf{E}}_A) d\mathbf{X}, \quad (2.38)$$

so comparison with (2.34) gives

$$\mathbf{F} = F_{iA} \mathbf{e}_i \otimes \hat{\mathbf{E}}_A, \quad F_{iA} = \chi_{i,A}. \quad (2.39)$$

The mixed basis $\{\mathbf{e}_i \otimes \hat{\mathbf{E}}_A\}$ shows $\mathbf{F}$ is a *two-point tensor*, a linear map $\hat{\mathbb{E}}^3 \rightarrow \mathbb{E}^3$ (exactly the object of Problem 12 of Chapter 1). Its action on a reference basis vector,

$$\mathbf{F} \hat{\mathbf{E}}_B = (F_{iA} \mathbf{e}_i \otimes \hat{\mathbf{E}}_A) \hat{\mathbf{E}}_B = F_{iA} (\hat{\mathbf{E}}_A \cdot \hat{\mathbf{E}}_B) \mathbf{e}_i = F_{iA} \delta_{AB} \mathbf{e}_i = F_{iB} \mathbf{e}_i, \quad (2.40)$$

gives the component extraction $F_{iA} = \mathbf{e}_i \cdot \mathbf{F} \hat{\mathbf{E}}_A$ (2.41).

LEMMA 2.15 (product of two dyads). *For all vectors, $(\mathbf{a} \otimes \mathbf{b})(\mathbf{c} \otimes \mathbf{d}) = (\mathbf{b} \cdot \mathbf{c}) \mathbf{a} \otimes \mathbf{d}$.*

Proof. Act on an arbitrary $\mathbf{v}$ and unfold using $(\mathbf{c} \otimes \mathbf{d})\mathbf{v} = (\mathbf{d} \cdot \mathbf{v})\mathbf{c}$ and $(\mathbf{a} \otimes \mathbf{b})\mathbf{c} = (\mathbf{b} \cdot \mathbf{c})\mathbf{a}$:

$$\begin{aligned} [(\mathbf{a} \otimes \mathbf{b})(\mathbf{c} \otimes \mathbf{d})]\mathbf{v} &= (\mathbf{a} \otimes \mathbf{b})[(\mathbf{d} \cdot \mathbf{v})\mathbf{c}] = (\mathbf{d} \cdot \mathbf{v})(\mathbf{b} \cdot \mathbf{c})\mathbf{a} \\ &= (\mathbf{b} \cdot \mathbf{c})(\mathbf{d} \cdot \mathbf{v})\mathbf{a} = (\mathbf{b} \cdot \mathbf{c})(\mathbf{a} \otimes \mathbf{d})\mathbf{v}, \end{aligned} \quad (2.43-2.44)$$

true for all $\mathbf{v}$, so the tensors are equal. $\blacksquare$

Applying Lemma 2.15 to (2.37) with $\text{grad } \mathbf{u} = \tilde{u}_{i,k} \mathbf{e}_i \otimes \mathbf{e}_k$ and $\mathbf{F} = F_{jA} \mathbf{e}_j \otimes \hat{\mathbf{E}}_A$,

$$\begin{aligned} \text{Grad } \mathbf{u} &= (\tilde{u}_{i,k} \mathbf{e}_i \otimes \mathbf{e}_k)(F_{jA} \mathbf{e}_j \otimes \hat{\mathbf{E}}_A) = \tilde{u}_{i,k} F_{jA} (\mathbf{e}_k \cdot \mathbf{e}_j) \mathbf{e}_i \otimes \hat{\mathbf{E}}_A \\ &= \tilde{u}_{i,k} x_{j,A} \delta_{kj} \mathbf{e}_i \otimes \hat{\mathbf{E}}_A = \tilde{u}_{i,j} x_{j,A} \mathbf{e}_i \otimes \hat{\mathbf{E}}_A = \hat{u}_{i,A} \mathbf{e}_i \otimes \hat{\mathbf{E}}_A, \end{aligned} \quad (2.42)$$

the last step being the ordinary chain rule $\hat{u}_{i,A} = \tilde{u}_{i,j} x_{j,A}$. So $\text{Grad } \underline{u}$ is a two-point tensor too.

Material curves. Let $\underline{X}(S)$, S arclength, parametrise a curve $C \subset \kappa$ of fixed material points; at time t^* it is convected to $\underline{x}_*(S) = \chi_\kappa(\underline{X}(S), t^*)$ (Figure 2.7). The relative position of two of its points is obtained by integrating the tangent,

$$\begin{aligned} \underline{x}_*(S_1) - \underline{x}_*(S_0) &= \underline{e}_i \int_{S_0}^{S_1} x'_{*i}(S) \, dS \\ &= \underline{e}_i \int_{S_0}^{S_1} F_{iA}(X_B(S), t^*) X'_A(S) \, dS, \end{aligned} \quad (2.45\text{--}2.46)$$

using $x'_{*i} = \chi_{i,A} X'_A = F_{iA} X'_A$: knowing $\underline{F}$ lets us reconstruct relative positions of connected material points.

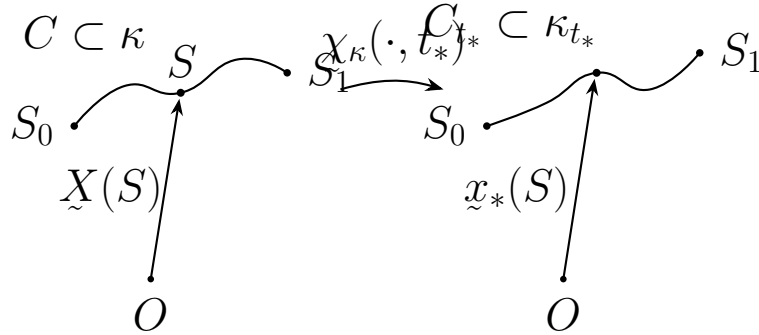


Figure 2.7: A material curve and its image in κ_{t^*} .

REMARK 2.16 (Figure 2.7). Figure 2.7 shows a curve C of fixed material points in κ and its deformed image C_{t^*} in κ_{t^*} ; the arclength S labels the material points and rides along with them. The tangent transforms by $\underline{F}$, $\underline{x}'_* = \underline{F} \underline{X}'$, the seed of the stretch formula $\underline{\mu m} = \underline{F} \underline{M}$ of §2.6.

Volume. Invertibility of χ_κ at each t makes $\underline{F}$ invertible (inverse function theorem), and invertibility of a tensor is $\det \underline{F} \neq 0$, where, from Chapter 1,

$$\det \underline{F} = \frac{[\underline{F} \underline{a}, \underline{F} \underline{b}, \underline{F} \underline{c}]}{[\underline{a}, \underline{b}, \underline{c}]} \quad (2.47)$$

for any independent $\{\underline{a}, \underline{b}, \underline{c}\}$ (well defined for the two-point $\underline{F}$ because numerator and denominator each use vectors of a single space).

REMARK 2.17 (invertibility of $\underline{F}$ is only *local* information). The implication runs one way only. Global invertibility of the motion $\underline{\chi}_\kappa(\cdot, t)$ gives, by the inverse function theorem, an inverse $\underline{F}^{-1}$ at each $\underline{X}$. But the existence of $\underline{F}^{-1}$ at a *particular* $\underline{X}$ does *not* imply that $\underline{\chi}_\kappa$ is invertible at all $\underline{X}$: it implies only that $\underline{\chi}_\kappa$ is invertible in some *neighbourhood* of the point considered. Nothing local can prevent a motion from folding a distant part of the body onto the piece near $\underline{X}$; $\det \underline{F} \neq 0$ everywhere forbids local self-penetration, not global. This is the same phenomenon as $x \mapsto e^{ix}$ having nonzero derivative everywhere while being far from injective, and it is why global injectivity of the motion is imposed as a separate physical assumption in Definition 2.2 rather than deduced from $\det \underline{F} > 0$.

Three material curves through $\underline{X}$ carry tangents $d\underline{X}, d\underline{Y}, d\underline{Z}$ spanning a reference parallelepiped of volume $dV = [d\underline{X}, d\underline{Y}, d\underline{Z}]$ (Figure 2.8); their images $d\underline{x} = \underline{F}d\underline{X}$ etc. span $d\underline{v} = [d\underline{x}, d\underline{y}, d\underline{z}]$. Taking $\{\underline{a}, \underline{b}, \underline{c}\} = \{d\underline{X}, d\underline{Y}, d\underline{Z}\}$ in (2.47),

$$\begin{aligned} d\underline{v} &= [\underline{F}d\underline{X}, \underline{F}d\underline{Y}, \underline{F}d\underline{Z}] = \det \underline{F} [d\underline{X}, d\underline{Y}, d\underline{Z}] \\ &= J dV, \quad J = \det \underline{F} > 0. \end{aligned} \tag{2.48–2.50}$$

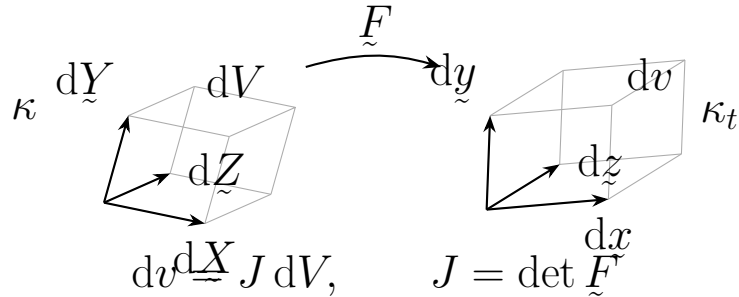


Figure 2.8: Three material curves intersecting at a material point.

REMARK 2.18 (Figure 2.8, and J as a volume ratio). Figure 2.8 shows the reference parallelepiped (left, κ) and its image (right, κ_t): $J =$

$\det \underline{\underline{F}}$ is the factor by which local volume is scaled, exactly the “oriented volume ratio” reading of the determinant from Chapter 1. If the reference configuration is *occupiable* (could be physically occupied), then $J > 0$; if not, one only knows $J = |\det \underline{\underline{F}}|$. Incompressibility is the constraint $J \equiv 1$.

Area (the Piola–Nanson formula). Let $d\underline{\underline{X}}, d\underline{\underline{Y}}$ span the tangent plane of a material surface at $\underline{\underline{X}}$, of area $dA = |d\underline{\underline{X}} \times d\underline{\underline{Y}}|$, with unit normal $\underline{\underline{N}}$ defined by

$$\underline{\underline{N}} dA = d\underline{\underline{X}} \times d\underline{\underline{Y}} \quad (2.51)$$

(Figure 2.9). The deformed edges $d\underline{\underline{x}} = \underline{\underline{F}}d\underline{\underline{X}}, d\underline{\underline{y}} = \underline{\underline{F}}d\underline{\underline{Y}}$ span a parallelogram of area da with unit normal $\underline{\underline{n}}, \underline{\underline{n}} da = d\underline{\underline{x}} \times d\underline{\underline{y}}$. Using the cofactor $\underline{\underline{F}}^*$, defined (as in Chapter 1) by $\underline{\underline{F}}^*(\underline{\underline{a}} \times \underline{\underline{b}}) = \underline{\underline{F}}\underline{\underline{a}} \times \underline{\underline{F}}\underline{\underline{b}}$,

$$\underline{\underline{n}} da = \underline{\underline{F}}d\underline{\underline{X}} \times \underline{\underline{F}}d\underline{\underline{Y}} = \underline{\underline{F}}^*(d\underline{\underline{X}} \times d\underline{\underline{Y}}) = \underline{\underline{F}}^*\underline{\underline{N}} dA, \quad (2.52\text{--}2.54)$$

the *Piola–Nanson formula*, in components $n_i da = F_{iB}^* N_B dA$. By Problem 22(c) of Chapter 1, $\underline{\underline{A}}^t \underline{\underline{A}}^* = (\det \underline{\underline{A}})\underline{\underline{I}}$, valid also for the two-point $\underline{\underline{F}}$, so

$$\underline{\underline{F}}^* = J \underline{\underline{F}}^{-t}. \quad (2.55)$$

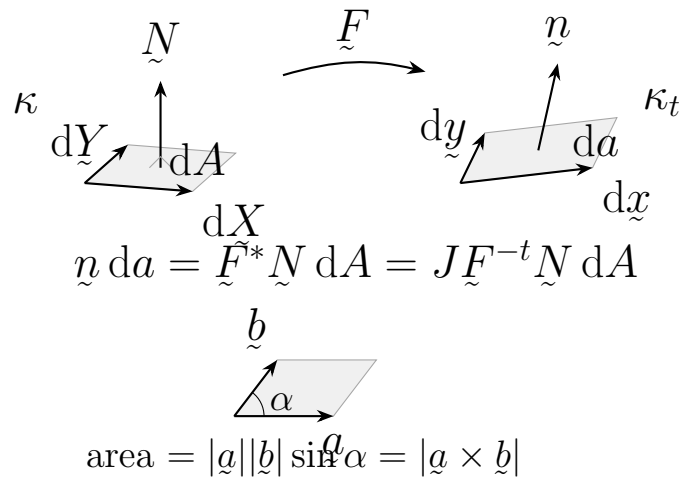


Figure 2.9: Deformation of oriented surface measure.

REMARK 2.19 (Figure 2.9: area is not a passive rider). Figure 2.9 shows the reference patch with edges $d\tilde{X}, d\tilde{Y}$, area dA , normal $\tilde{N}$, and its deformed image with $d\tilde{x}, d\tilde{y}$, area da , normal $\tilde{n}$; the small inset recalls $\text{area} = |\underline{a}||\underline{b}|\sin\alpha = |\underline{a} \times \underline{b}|$ from Chapter 1. The point of $\tilde{n} da = J\tilde{F}^{-t}\tilde{N} dA$ is that the normal does *not* transform by $\tilde{F}$ (that would be the tangent's rule): it transforms by the cofactor $\tilde{F}^* = J\tilde{F}^{-t}$, because a normal is defined by a cross product, and cross products go with cofactors. This single formula converts every surface traction and flux between reference and current descriptions in the balance laws.

2.5 Mass and density

Consider a sub-body $S \subset B$, occupying a part $P = \kappa(S) \subset \kappa$ of the reference configuration and a part $P_t = \chi_\kappa(P, t) \subset \kappa_t$ of the current one. Mass is assumed absolutely continuous in volume (more volume, more mass) and strictly positive on regions of positive volume, so there are positive *density* fields $\rho(\tilde{x}, t)$ and $\rho_\kappa(\tilde{X}, t)$ with

$$m(S, t) = \int_{P_t} \rho(\tilde{x}, t) dv = \int_P \rho_\kappa(\tilde{X}, t) dV. \quad (2.56)$$

(Conservation of mass, treated later, will make m independent of t .)

Change variables in the first integral by $\tilde{x} = \chi_\kappa(\tilde{X}, t)$, $dv = J dV$ from (2.50):

$$\int_{P_t} \rho(\tilde{x}, t) dv = \int_P \rho(\chi_\kappa(\tilde{X}, t), t) J(\tilde{X}, t) dV, \quad (2.57)$$

so equality of the two forms in (2.56) becomes

$$\int_P [\rho_\kappa(\tilde{X}, t) - \rho(\chi_\kappa(\tilde{X}, t), t) J(\tilde{X}, t)] dV = 0 \quad \text{for all } P \subset \kappa. \quad (2.58)$$

It is *sufficient* for this that the bracket vanish pointwise,

$$\rho_\kappa = \rho J \quad \text{at every } \tilde{X} \in \kappa. \quad (2.59)$$

That it is also *necessary*, when the integrand is continuous, is the content of the localization theorem.

THEOREM 2.20 (Localization). *Let $\varphi(\tilde{X})$ be continuous and suppose*

$$\int_P \varphi(\tilde{X}) \, dV = 0 \quad \text{for all } P \subset \kappa. \quad (2.60)$$

Then

$$\varphi(\tilde{X}) = 0 \quad \text{at every } \tilde{X} \in \kappa. \quad (2.61)$$

(The converse statement is obvious: if φ vanishes identically its integral over every P is zero.)

Proof. Fix $\tilde{X}_0 \in \kappa$ and take $P = R_\varepsilon$, the ball of radius ε and volume V_ε centred at $\tilde{X}_0$ and contained in κ . Consider the average discrepancy

$$I_\varepsilon = \left| \frac{1}{V_\varepsilon} \int_{R_\varepsilon} [\varphi(\tilde{X}_0) - \varphi(\tilde{X})] \, dV \right|. \quad (2.62)$$

By the triangle inequality for integrals, then bounding the integrand by its maximum over the (compact) ball,

$$\begin{aligned} I_\varepsilon &\leq \frac{1}{V_\varepsilon} \int_{R_\varepsilon} |\varphi(\tilde{X}_0) - \varphi(\tilde{X})| \, dV \\ &\leq \frac{1}{V_\varepsilon} \int_{R_\varepsilon} \max_{\tilde{X} \in R_\varepsilon} |\varphi(\tilde{X}_0) - \varphi(\tilde{X})| \, dV \\ &= \max_{\tilde{X} \in R_\varepsilon} |\varphi(\tilde{X}_0) - \varphi(\tilde{X})|, \end{aligned} \quad (2.63)$$

the last step because the maximum is a constant and $\int_{R_\varepsilon} dV = V_\varepsilon$. A continuous function on a compact set attains its maximum, and by continuity of φ that maximum tends to 0 as the ball shrinks:

$$0 \leq I_\varepsilon \leq \max_{\tilde{X} \in R_\varepsilon} |\varphi(\tilde{X}_0) - \varphi(\tilde{X})| \longrightarrow 0 \quad (\varepsilon \rightarrow 0). \quad (2.64)$$

But the hypothesis (2.60) makes $\int_{R_\varepsilon} \varphi(\tilde{X}) \, dV = 0$, so

$$\varphi(\tilde{X}_0) = \frac{1}{V_\varepsilon} \int_{R_\varepsilon} \varphi(\tilde{X}_0) \, dV = \frac{1}{V_\varepsilon} \int_{R_\varepsilon} [\varphi(\tilde{X}_0) - \varphi(\tilde{X})] \, dV \xrightarrow{\varepsilon \rightarrow 0} 0, \quad (2.65)$$

i.e. $|\varphi(\tilde{X}_0)| = \lim_{\varepsilon \rightarrow 0} I_\varepsilon = 0$. Since $\tilde{X}_0$ was arbitrary, $\varphi \equiv 0$. ■

Applying the theorem to the continuous bracket in (2.58) yields the pointwise mass relation (2.59), $\rho_\kappa = \rho J$. Continuity of that bracket is guaranteed by the assumed differentiability of χ_κ (which makes $\tilde{F}$, and hence $J = \det \tilde{F}$, continuous) together with continuity of ρ .

REMARK 2.21 (what localization really says). The theorem is the bridge from *integral* (global) balance laws to *pointwise* (local) field equations, and it is used again and again: conservation of mass, momentum, and energy are all first stated as “the integral over every sub-body vanishes”, and localization converts each into a differential equation holding at every point. Intuitively, if an average over *every* shrinking ball around $\underline{X}_0$ is forced to be zero, the value at $\underline{X}_0$ cannot be anything but zero, since a continuous function is nearly constant on a small enough ball and its average there nearly equals its central value. The relation $\rho_\kappa = \rho J$ itself is just conservation of mass in disguise: the same matter in a smaller current volume ($J < 1$) must be denser, $\rho = \rho_\kappa / J$.

2.6 Material curves and the shifter

Return to the material curve $C \subset \kappa$ and its convected image $C_{t_*} = \chi_\kappa(C, t_*) \subset \kappa_{t_*}$ (Figure 2.10). The material point at arclength station $\underline{S}$ on C occupies $\underline{x}_*(S) = \chi_\kappa(\underline{X}(S), t_*)$. Differentiating in S at fixed t_* , by the chain rule and (2.34),

$$\underline{x}'_*(S) = \tilde{F}(\underline{X}(S), t_*) \underline{M}(S), \quad \underline{M}(S) = \underline{X}'(S), \quad (2.66)$$

$\underline{M}$ being a *unit* tangent to C (because S is arclength, $|\underline{X}'| = 1$). The image $\underline{x}'_*$ is tangent to C_{t_*} but need not be a unit vector; writing $\underline{x}'_* = \mu \underline{m}$ with $|\underline{m}| = 1$ and $\mu = |\underline{x}'_*|$ the local *stretch*,

$$\mu \underline{m} = \tilde{F} \underline{M}. \quad (2.67)$$

REMARK 2.22 (Figure 2.10: what $\tilde{F}$ does to a fibre). Figure 2.10 shows the curve $C \subset \kappa$ with station S , position $\underline{X}(S)$ and unit tangent $\underline{M}$, beside its image $C_{t_*} \subset \kappa_{t_*}$ with $\underline{x}_*(S) = \chi_\kappa(\underline{X}(S), t_*)$. Equation (2.67)

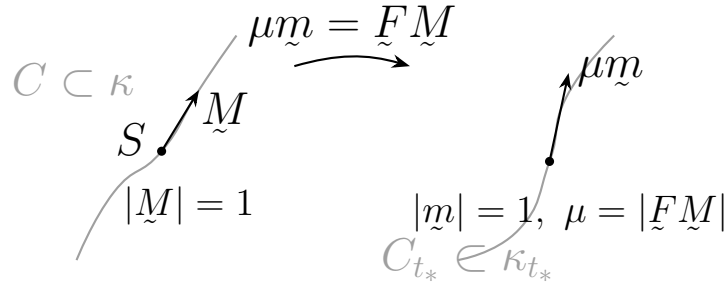


Figure 2.10: Orientation of material curves at point p with arclength S in κ .

is the sentence “ $\tilde{F}$ takes the unit material fibre $\tilde{M}$ and returns the deformed fibre, whose length μ is the stretch and whose direction $\tilde{m}$ is the new orientation”. Every deformation measure in §2.7 is extracted from this one statement by taking lengths and angles of $\tilde{F}\tilde{M}$.

Interpreting (2.67) for an *undeformed* body raises a genuine conceptual difficulty. We stipulated $\tilde{X} \in \hat{\mathbb{E}}^3$ but $\kappa_t \subset \mathcal{E}^3$ for every t , including t_0 ; so if we want to say “at t_0 the body is where its reference copy is”, we are comparing vectors from two different spaces, which is meaningless. The repair is a fixed two-point tensor that carries one space onto the other.

DEFINITION 2.23 (The shifter). Introduce fixed orthonormal vectors $\tilde{E}_A \in \mathbb{E}^3$ and the uniform two-point tensor

$$\underline{1} = \tilde{E}_A \otimes \hat{E}_A. \quad (2.68)$$

If $\{\hat{E}_A\}$ and $\{\tilde{E}_A\}$ have the same handedness, $\underline{1}$ rotates the former onto the latter and is a two-point rotation; otherwise it is merely orthogonal. Resolving $\tilde{E}_A \in \mathbb{E}^3$ on $\{\tilde{e}_i\}$ by the orthonormal expansion of Chapter 1,

$$\tilde{E}_A = \delta_{iA} \tilde{e}_i, \quad \delta_{iA} = \tilde{e}_i \cdot \tilde{E}_A, \quad (2.69)$$

(the dot product is legitimate: both factors lie in $\mathbb{E}^3$), giving the standard two-point decomposition

$$\underline{1} = \delta_{iA} \tilde{e}_i \otimes \hat{E}_A. \quad (2.70)$$

The δ_{iA} are Kronecker deltas if and only if $\{\underline{E}_A\} = \{\underline{e}_i\}$; otherwise they are the cosines of the fixed angles between the two sets.

PROPOSITION 2.24 (Algebra of the shifter). *With $\underline{I}$ and $\hat{\underline{I}}$ the identities of $\mathbb{E}^3$ and $\hat{\mathbb{E}}^3$,*

$$\underline{1} = \underline{1}\hat{\underline{I}} = \underline{I}\underline{1}, \quad \underline{1}^t \underline{1} = \hat{\underline{I}}, \quad \underline{1}\underline{1}^t = \underline{I}, \quad (2.71-2.73)$$

equivalently, in components,

$$\delta_{iA}\delta_{iB} = \delta_{AB}, \quad \delta_{iA}\delta_{jA} = \delta_{ij}. \quad (2.74)$$

Proof. For (2.71): $\underline{E}_A = \underline{I}\underline{E}_A$ and $\hat{\underline{E}}_A = \hat{\underline{I}}\hat{\underline{E}}_A$, so by the rules $\underline{A}(\underline{a} \otimes \underline{b}) = (\underline{A}\underline{a}) \otimes \underline{b}$ and $(\underline{a} \otimes \underline{b})\underline{A} = \underline{a} \otimes (\underline{A}^t \underline{b})$ (Problems 17(b),(c) of Chapter 1),

$$\begin{aligned} \underline{I}\underline{1} &= \underline{I}(\underline{E}_A \otimes \hat{\underline{E}}_A) = (\underline{I}\underline{E}_A) \otimes \hat{\underline{E}}_A = \underline{1}, \\ \underline{1}\hat{\underline{I}} &= \underline{E}_A \otimes (\hat{\underline{I}}^t \hat{\underline{E}}_A) = \underline{E}_A \otimes \hat{\underline{E}}_A = \underline{1}. \end{aligned}$$

For (2.72), transpose (2.68) by Problem 17(a) of Chapter 1, which gives $(\underline{a} \otimes \underline{b})^t = \underline{b} \otimes \underline{a}$, and then multiply using Lemma 2.15:

$$\begin{aligned} \underline{1}^t \underline{1} &= (\hat{\underline{E}}_A \otimes \underline{E}_A)(\underline{E}_B \otimes \hat{\underline{E}}_B) = (\underline{E}_A \cdot \underline{E}_B) \hat{\underline{E}}_A \otimes \hat{\underline{E}}_B \\ &= \delta_{AB} \hat{\underline{E}}_A \otimes \hat{\underline{E}}_B = \hat{\underline{E}}_A \otimes \hat{\underline{E}}_A = \hat{\underline{I}}. \end{aligned} \quad (2.72)$$

Identically, $\underline{1}\underline{1}^t = (\hat{\underline{E}}_A \cdot \hat{\underline{E}}_B)\underline{E}_A \otimes \underline{E}_B = \underline{E}_A \otimes \underline{E}_A = \underline{I}$. In components, (2.72) reads $\delta_{iA}\delta_{iB} = \delta_{AB}$ and (2.73) reads $\delta_{iA}\delta_{jA} = \delta_{ij}$, which is (2.74). ■

Acting with the shifter on (2.23) produces a *spatial* copy of the reference position,

$$\underline{1}X = X_A \underline{1}\hat{\underline{E}}_A = X_A \underline{E}_A \in \mathbb{E}^3, \quad (2.75)$$

and the statement $\kappa = \kappa_{t_0}$ is then properly written

$$\chi_\kappa(X, t_0) = \underline{1}X \quad (2.76)$$

(taking the origins of Figure 2.6 to coincide, without loss of generality).

This makes the *displacement* well defined:

$$\underline{u}(X, t) = \chi_\kappa(X, t) - \underline{1}X \in \mathbb{E}^3, \quad \hat{u}_i(X_B, t) = \chi_i(X_B, t) - \delta_{iA}X_A. \quad (2.77-2.78)$$

REMARK 2.25 (why the shifter is not pedantry). The definition of displacement found in most of the literature omits $\underline{1}$ and writes $\underline{u} = \underline{\chi}_\kappa - \underline{X}$. That expression subtracts a vector of $\mathbb{E}^3$ from a vector of $\hat{\mathbb{E}}^3$: an impermissible operation between different vector spaces, and a documented source of confusion in using displacement to characterise deformation. The shifter is the “two-point identity tensor”: when $\underline{E}_A$ and $\hat{\underline{E}}_A$ are aligned it simply *parallel transports* a vector from one copy of space to the other and changes nothing else, so its components δ_{iA} then really are Kronecker deltas and every classical formula is recovered. Its only job is to make the bookkeeping honest, exactly as the caret did for functions (Remark 2.6). The shifter and the displacement are put to substantive use later, in the book’s Chapters 7 and 12.

PROPOSITION 2.26 (Undeformed state). *If (2.76) holds then*

$$\underline{F}(\underline{X}, t_0) = \underline{1}, \quad F_{iA} = \delta_{iA}, \quad (2.79)$$

and consequently $\mu = 1$ for every material curve:

$$\underline{\mu} \underline{m} = \underline{1} \underline{M}, \quad \mu = \sqrt{\underline{1} \underline{M} \cdot \underline{1} \underline{M}} = \sqrt{\underline{M} \cdot \underline{M}} = 1. \quad (2.80\text{--}2.81)$$

Moreover, in the general case (2.67) forces $\mu > 0$.

Proof. (2.79): $\underline{F} = \text{Grad } \underline{\chi}_\kappa$ and $\underline{\chi}_\kappa(\underline{X}, t_0) = \underline{1} \underline{X}$ is *linear* in $\underline{X}$ with the uniform (position independent) coefficient $\underline{1}$, so its gradient is $\underline{1}$ itself; in components $F_{iA} = \partial(\delta_{jB} X_B) / \partial X_A \big|_{i=j} = \delta_{iA}$. Substituting into (2.67) gives (2.80), and since the shifter is orthogonal, $\underline{1} \underline{M} \cdot \underline{1} \underline{M} = \underline{M} \cdot \underline{1}^t \underline{1} \underline{M} = \underline{M} \cdot \hat{\underline{I}} \underline{M} = \underline{M} \cdot \underline{M} = 1$ by (2.72), so $\mu = 1$: an undeformed body stretches no fibre. Finally $\mu > 0$ always, for $\mu = 0$ would give $\underline{F} \underline{M} = \underline{0}$ with $\underline{M} \neq \underline{0}$ (as $|\underline{M}| = 1$), contradicting invertibility of $\underline{F}$. ■

2.7 The Cauchy–Green deformation tensors and the Lagrange–Euler strain tensors

The stretch is the positive square root of μ^2 , and (2.67) turns that into a quadratic form in $\underline{M}$. Using only the definitions of the transpose and

of the product of two tensors (Chapter 1),

$$\mu^2 = |\mu \underline{m}|^2 = \underline{F} \underline{M} \cdot \underline{F} \underline{M} = \underline{M} \cdot \underline{F}^t(\underline{F} \underline{M}) = \underline{M} \cdot (\underline{F}^t \underline{F}) \underline{M}, \quad (2.82)$$

so that

$$0 < \mu^2 = \underline{M} \cdot \underline{C} \underline{M}, \quad \underline{C} = \underline{F}^t \underline{F} \quad (2.83)$$

defines the *right Cauchy–Green deformation tensor*.

PROPOSITION 2.27 (Properties of $\underline{C}$). $\underline{C}$ is a referential tensor ($\hat{\mathbb{E}}^3 \rightarrow \hat{\mathbb{E}}^3$), symmetric, and positive definite; its components are

$$\begin{aligned} \underline{C} &= C_{AB} \hat{\underline{E}}_A \otimes \hat{\underline{E}}_B, & C_{AB} &= F_{iA} F_{iB} = C_{BA}, \\ \mu^2 &= C_{AB} M_A M_B. \end{aligned} \quad (2.86\text{--}2.88)$$

Proof. Domain. $\underline{F} : \hat{\mathbb{E}}^3 \rightarrow \mathbb{E}^3$, hence $\underline{F}^t : \mathbb{E}^3 \rightarrow \hat{\mathbb{E}}^3$, hence $\underline{C} = \underline{F}^t \underline{F}$ maps $\hat{\mathbb{E}}^3$ into itself. *Positive definiteness.* For any $\underline{a} \neq \underline{0}$ in $\hat{\mathbb{E}}^3$, moving $\underline{F}^t$ across the dot product by the definition of the transpose,

$$\underline{a} \cdot \underline{C} \underline{a} = \underline{a} \cdot \underline{F}^t \underline{F} \underline{a} = \underline{F} \underline{a} \cdot \underline{F} \underline{a} = |\underline{F} \underline{a}|^2 > 0, \quad (2.84)$$

strictly, because $\underline{F}$ is invertible ($\det \underline{F} \neq 0$) so $\underline{F} \underline{a} \neq \underline{0}$. *Symmetry*, using $(\underline{B} \underline{A})^t = \underline{A}^t \underline{B}^t$ (Problem 20 of Chapter 1) and $(\underline{F}^t)^t = \underline{F}$:

$$\underline{C}^t = (\underline{F}^t \underline{F})^t = \underline{F}^t (\underline{F}^t)^t = \underline{F}^t \underline{F} = \underline{C}. \quad (2.85)$$

Components. With $\underline{F} = F_{iA} \underline{e}_i \otimes \hat{\underline{E}}_A$ and $\underline{F}^t = F_{iA} \hat{\underline{E}}_A \otimes \underline{e}_i$ (Problem 17(a) of Chapter 1), multiply by Lemma 2.15:

$$\begin{aligned} \underline{C} &= (F_{iA} \hat{\underline{E}}_A \otimes \underline{e}_i)(F_{jB} \underline{e}_j \otimes \hat{\underline{E}}_B) = F_{iA} F_{jB} (\underline{e}_i \cdot \underline{e}_j) \hat{\underline{E}}_A \otimes \hat{\underline{E}}_B \\ &= F_{iA} F_{jB} \delta_{ij} \hat{\underline{E}}_A \otimes \hat{\underline{E}}_B = C_{AB} \hat{\underline{E}}_A \otimes \hat{\underline{E}}_B, & C_{AB} &= F_{iA} F_{iB}, \end{aligned} \quad (2.86\text{--}2.87)$$

manifestly symmetric since $F_{iA} F_{iB} = F_{iB} F_{iA}$. Finally

$$\mu^2 = \underline{M} \cdot (C_{AB} \hat{\underline{E}}_A \otimes \hat{\underline{E}}_B) \underline{M} = \underline{M} \cdot C_{AB} M_B \hat{\underline{E}}_A = C_{AB} M_A M_B, \quad (2.88)$$

so (C_{AB}) is a positive definite matrix. ■

Inverting (2.67) gives the companion, spatial, measure. Applying F^{-1} to $\mu \underline{m} = \underline{F} M$,

$$\begin{aligned} \mu \underline{F}^{-1} \underline{m} &= \underline{F}^{-1}(\underline{F} M) = (\underline{F}^{-1} \underline{F}) M = \hat{I} M = M, \\ \text{i.e. } \mu^{-1} M &= \underline{F}^{-1} \underline{m}, \end{aligned} \quad (2.89\text{--}2.90)$$

and taking the squared length of the left side (recall $|M| = 1$),

$$\begin{aligned} \mu^{-2} &= |\mu^{-1} M|^2 = \underline{F}^{-1} \underline{m} \cdot \underline{F}^{-1} \underline{m} \\ &= \underline{m} \cdot (\underline{F}^{-1})^t \underline{F}^{-1} \underline{m} = \underline{m} \cdot \underline{F}^{-t} \underline{F}^{-1} \underline{m} = \underline{m} \cdot \underline{B}^{-1} \underline{m}, \end{aligned} \quad (2.91)$$

where

$$\underline{B} = \underline{F} \underline{F}^t \quad (2.92)$$

is the *left Cauchy–Green deformation tensor*; indeed $\underline{B}^{-1} = (\underline{F} \underline{F}^t)^{-1} = \underline{F}^{-t} \underline{F}^{-1}$ by Problems 18–19 of Chapter 1.

PROPOSITION 2.28 (Properties of $\underline{B}$). *$\underline{B}$ is a spatial tensor ($\mathbb{E}^3 \rightarrow \mathbb{E}^3$), symmetric, with*

$$\underline{B} = B_{ij} \underline{e}_i \otimes \underline{e}_j, \quad B_{ij} = F_{iA} F_{jA} = B_{ji}, \quad (2.93\text{--}2.94)$$

and $\underline{B}^{-1}$ is positive definite.

Proof. $\underline{F}^t : \mathbb{E}^3 \rightarrow \hat{\mathbb{E}}^3$ then $\underline{F} : \hat{\mathbb{E}}^3 \rightarrow \mathbb{E}^3$, so $\underline{B}$ maps $\mathbb{E}^3$ to itself; symmetry is $\underline{B}^t = (\underline{F} \underline{F}^t)^t = \underline{F} \underline{F}^t = \underline{B}$. Components, again by Lemma 2.15:

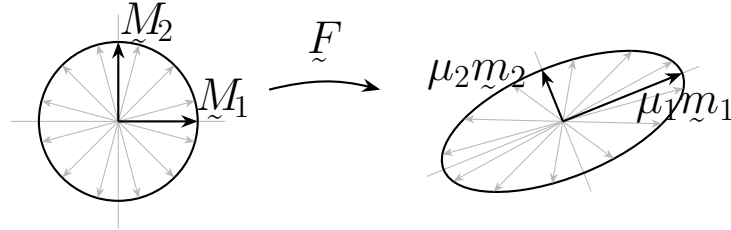
$$\begin{aligned} \underline{B} &= (F_{iA} \underline{e}_i \otimes \hat{\underline{E}}_A)(F_{jB} \hat{\underline{E}}_B \otimes \underline{e}_j) = F_{iA} F_{jB} (\hat{\underline{E}}_A \cdot \hat{\underline{E}}_B) \underline{e}_i \otimes \underline{e}_j \\ &= F_{iA} F_{jB} \delta_{AB} \underline{e}_i \otimes \underline{e}_j = B_{ij} \underline{e}_i \otimes \underline{e}_j, \quad B_{ij} = F_{iA} F_{jA}. \end{aligned} \quad (2.93\text{--}2.94)$$

Positive definiteness of $\underline{B}^{-1}$: for $\underline{a} \neq \underline{0}$ in $\mathbb{E}^3$,

$$\underline{a} \cdot \underline{B}^{-1} \underline{a} = \underline{a} \cdot \underline{F}^{-t} \underline{F}^{-1} \underline{a} = \underline{F}^{-1} \underline{a} \cdot \underline{F}^{-1} \underline{a} = |\underline{F}^{-1} \underline{a}|^2 > 0, \quad (2.95)$$

since $\underline{F}^{-1}$ is invertible (with inverse $\underline{F}$). In Chapter 3 this is shown equivalent to positive definiteness of $\underline{B}$ itself. ■

REMARK 2.29 ($\underline{\mathcal{C}}$ and $\underline{\mathcal{B}}$: which space you measure in). $\underline{F}$ has one foot in each configuration, so it cannot be symmetric and carries rotation as well as stretch. Pairing $\underline{F}$ with its transpose cancels the rotation and lands the result wholly in one space: $\underline{\mathcal{C}} = \underline{F}^t \underline{F}$ lives in the *reference* space and is fed reference fibres $\underline{M}$, while $\underline{\mathcal{B}} = \underline{F} \underline{F}^t$ lives in the *current* space and is fed current fibres $\underline{m}$. Concretely, (2.83) says “give me the undeformed direction and I return the square of the stretch”, whereas (2.91) says “give me the deformed direction and I return the inverse square of the stretch”. If $\underline{F}$ is a pure rotation, $\underline{F}^t \underline{F} = \hat{\underline{I}}$ and no fibre stretches, which is exactly why these products, and not $\underline{F}$, are the measures of *distortion*.



unit directions $|\underline{M}| = 1$ in κ images $\underline{F}\underline{M}$: an ellipse

SUPPLEMENTARY FIGURE B. What $\underline{\mathcal{C}} = \underline{F}^t \underline{F}$ measures. Feed $\underline{F}$ every unit direction $\underline{M}$ at a material point: the images $\underline{F}\underline{M}$ sweep out an *ellipse* (an ellipsoid in three dimensions), because $\mu^2 = \underline{M} \cdot \underline{\mathcal{C}} \underline{M}$ is a quadratic form in $\underline{M}$. The stretch μ is the length of the image, so it depends on direction; its largest and smallest values occur along the principal axes of the ellipse, which are the eigenvectors of $\underline{\mathcal{C}}$, the corresponding μ^2 being its eigenvalues. A general fibre is both stretched *and* rotated, and the change in angle between two fibres is the shear recorded by the off-diagonal $\underline{C}_{AB}$. A rigid motion sends the circle to a circle of equal radius, i.e. $\underline{\mathcal{C}} = \hat{\underline{I}}$.

REMARK 2.30 (reading Supplementary Figure B: why $\underline{F}$ splits). The picture already contains the polar decomposition theorem of Chapter 3. Turning the circle into the ellipse can be done in two stages: first stretch along a fixed pair of orthogonal directions of κ , then rotate the stretched figure into place. Symbolically $\underline{F} = \underline{R}\underline{U}$ with $\underline{U}$ symmetric positive definite (the stretch) and $\underline{R}$ orthogonal (the rotation). Squaring kills the rotation, exactly as the picture suggests:

$$\underline{\mathcal{C}} = \underline{F}^t \underline{F} = (\underline{R}\underline{U})^t (\underline{R}\underline{U}) = \underline{U}^t \underline{R}^t \underline{R} \underline{U} = \underline{U} \hat{\underline{I}} \underline{U} = \underline{U}^2,$$

using $(\underline{B}\underline{A})^t = \underline{A}^t \underline{B}^t$, $\underline{R}^t \underline{R} = \underline{\hat{I}}$ and $\underline{U}^t = \underline{U}$. So $\underline{C}$ records the ellipse's shape and size but nothing about its final orientation, which is the precise sense in which it measures distortion alone; $\mu^2 = \underline{M} \cdot \underline{C} \underline{M} = \underline{M} \cdot \underline{U}^2 \underline{M} = |\underline{U} \underline{M}|^2$ says the stretch is done entirely by $\underline{U}$. Symmetrically $\underline{B} = \underline{F} \underline{F}^t = \underline{R} \underline{U}^2 \underline{R}^t = \underline{V}^2$ with $\underline{V} = \underline{R} \underline{U} \underline{R}^t$, the same stretch seen from the current configuration.

DEFINITION 2.31 (Lagrange and Euler strains). The deviation of the stretch from its undeformed value 1 is measured by

$$\frac{1}{2}(\mu^2 - 1) = \frac{1}{2}(\underline{M} \cdot \underline{C} \underline{M} - 1) = \underline{M} \cdot \underline{E} \underline{M}, \quad \underline{E} = \frac{1}{2}(\underline{C} - \underline{\hat{I}}), \quad (2.96-2.97)$$

the *Lagrange strain tensor*, and by

$$\frac{1}{2}(1 - \mu^{-2}) = \underline{m} \cdot \underline{e} \underline{m}, \quad \underline{e} = \frac{1}{2}(\underline{I} - \underline{B}^{-1}), \quad (2.98-2.99)$$

the *Euler strain tensor*. Both are symmetric.

Verification. Since $|\underline{M}| = 1$, $\underline{M} \cdot \underline{\hat{I}} \underline{M} = \underline{M} \cdot \underline{M} = 1$, so $\underline{M} \cdot \underline{E} \underline{M} = \frac{1}{2}(\underline{M} \cdot \underline{C} \underline{M} - \underline{M} \cdot \underline{M}) = \frac{1}{2}(\mu^2 - 1)$, which is (2.96); the Euler case is identical using (2.91) and $|\underline{m}| = 1$. ■

REMARK 2.32 (why the factor $\frac{1}{2}$, and why strain at all). For $\mu \simeq 1$ write $\mu = 1 + \varepsilon$ with ε small; then $\mu^2 - 1 = (1 + \varepsilon)^2 - 1 = 2\varepsilon + \varepsilon^2 \simeq 2(\mu - 1)$. The factor $\frac{1}{2}$ is chosen precisely so that $\underline{M} \cdot \underline{E} \underline{M} \simeq \mu - 1$, the *extension* (change in length per unit original length), recovering the engineer's strain in the small-deformation limit; likewise $\underline{m} \cdot \underline{e} \underline{m} \simeq \mu - 1$. The reason for introducing strains at all, rather than working with $\underline{C}$ and $\underline{B}$, is that they *vanish* in the undeformed state: by (2.79), $\underline{F} = \underline{1}$ gives $\underline{C} = \underline{1}^t \underline{1} = \underline{\hat{I}}$ and $\underline{B} = \underline{1} \underline{1}^t = \underline{I}$ by (2.72)–(2.73), hence $\underline{E} = \underline{\hat{Q}}$ and $\underline{e} = \underline{Q}$. (Henceforth we drop the caret on the referential zero tensor and let context decide.) The names used here are not universal, and there are infinitely many equivalent strain measures; see Ogden's book for a survey.

REMARK 2.33 (symmetry is meaningless for two-point tensors). $\underline{E}$ and $\underline{e}$ are symmetric, but note carefully that symmetry and skewness are

defined only for tensors mapping a vector space *to itself*: $\tilde{A}^t = \pm \tilde{A}$ compares $\tilde{A}$ with an object of the same type. For a two-point tensor such as $\tilde{F}$, the transpose $\tilde{F}^t$ maps $\mathbb{E}^3 \rightarrow \hat{\mathbb{E}}^3$ while $\tilde{F}$ maps $\hat{\mathbb{E}}^3 \rightarrow \mathbb{E}^3$, so the equation $\tilde{F}^t = \pm \tilde{F}$ cannot even be written. Two-point tensors are therefore *neither* symmetric nor skew, which is another reason the deformation measures are built as $\tilde{F}^t \tilde{F}$ and $\tilde{F} \tilde{F}^t$.

Arclength of a deformed curve. On C , $d\tilde{X}(S) = \tilde{M} dS$ with $dS = |d\tilde{X}|$ (arclength parametrisation), so the deformed arclength measure is

$$ds = |d\tilde{x}| = \sqrt{\tilde{F}d\tilde{X} \cdot \tilde{F}d\tilde{X}} = \sqrt{\tilde{M} \cdot \tilde{C}\tilde{M}} dS = \mu(S, t_*) dS, \quad (2.100)$$

that is $ds/dS = \mu(S, t_*)$ (2.101), and the total length of the deformed curve, and its change, are

$$l(C_{t_*}) = \int_{S_0}^{S_1} \mu(S, t_*) dS, \quad l(C_{t_*}) - l(C) = \int_{S_0}^{S_1} [\mu(S, t_*) - 1] dS. \quad (2.102-2.103)$$

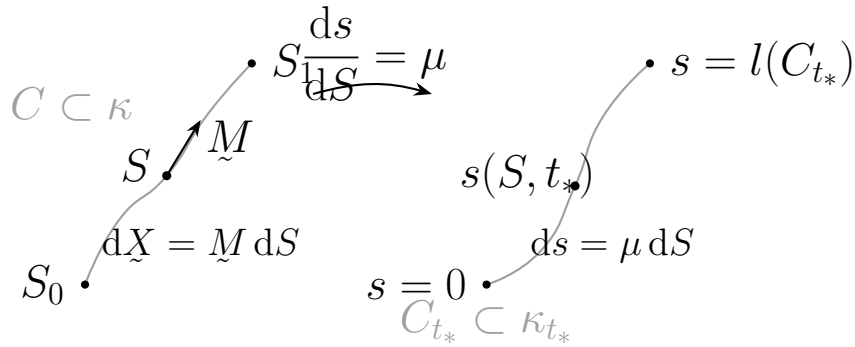


Figure 2.11: Arclength of a material point on material curves in κ and κ_{t_*} .

REMARK 2.34 (Figure 2.11). Figure 2.11 shows $C \subset \kappa$ carrying the stations S_0, S, S_1 , the unit tangent $\tilde{M}$ and the element $d\tilde{X} = \tilde{M} dS$, beside its image C_{t_*} carrying the deformed arclength $s(S, t_*)$. The content of (2.101) is that the stretch μ is literally the local ratio of deformed to undeformed arclength, so integrating it along the fibre gives the new length: a one-dimensional version of $dv = J dV$.

Angles between material curves. Let two material curves cross at $\tilde{X}$ with unit tangents $\tilde{M}_1, \tilde{M}_2$ and angle Θ ($\cos \Theta = \tilde{M}_1 \cdot \tilde{M}_2$); after deformation they cross at $\tilde{x}$ with unit tangents $\tilde{m}_1, \tilde{m}_2$ and angle θ ($\cos \theta = \tilde{m}_1 \cdot \tilde{m}_2$), where

$$\mu_1 \tilde{m}_1 = \tilde{F} \tilde{M}_1, \quad \mu_2 \tilde{m}_2 = \tilde{F} \tilde{M}_2 \quad (2.104)$$

(Figure 2.12). Dotting the two and moving $\tilde{F}^t$ across,

$$\mu_1 \mu_2 \cos \theta = \tilde{F} \tilde{M}_1 \cdot \tilde{F} \tilde{M}_2 = \tilde{M}_1 \cdot \tilde{C} \tilde{M}_2, \quad (2.105)$$

and dividing by $\mu_1 \mu_2 = \sqrt{\tilde{M}_1 \cdot \tilde{C} \tilde{M}_1} \sqrt{\tilde{M}_2 \cdot \tilde{C} \tilde{M}_2}$ from (2.83),

$$\cos \theta = \frac{\tilde{M}_1 \cdot \tilde{C} \tilde{M}_2}{\sqrt{(\tilde{M}_1 \cdot \tilde{C} \tilde{M}_1)(\tilde{M}_2 \cdot \tilde{C} \tilde{M}_2)}}, \quad (2.106)$$

in which $\tilde{M}_1 \cdot \tilde{C} \tilde{M}_2 = \tilde{M}_2 \cdot \tilde{C} \tilde{M}_1$ because $\tilde{M}_1 \cdot \tilde{C} \tilde{M}_2 = \tilde{C}^t \tilde{M}_1 \cdot \tilde{M}_2 = \tilde{M}_2 \cdot \tilde{C} \tilde{M}_1$ by the symmetry of $\tilde{C}$. Running the same argument backwards with $\mu_1^{-1} \tilde{M}_1 = \tilde{F}^{-1} \tilde{m}_1$, $\mu_2^{-1} \tilde{M}_2 = \tilde{F}^{-1} \tilde{m}_2$ from (2.90) gives the undeformed angle in terms of current data,

$$\cos \Theta = \frac{\tilde{m}_1 \cdot \tilde{B}^{-1} \tilde{m}_2}{\sqrt{(\tilde{m}_1 \cdot \tilde{B}^{-1} \tilde{m}_1)(\tilde{m}_2 \cdot \tilde{B}^{-1} \tilde{m}_2)}}. \quad (2.107-2.108)$$

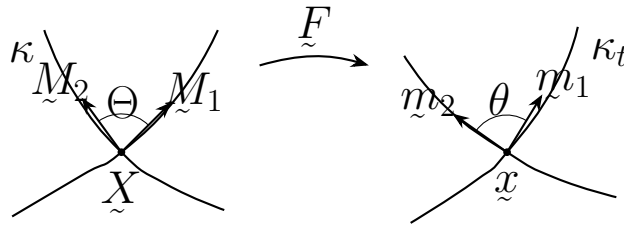


Figure 2.12: Intersecting material curves in κ and κ_{t*} .

REMARK 2.35 (Figure 2.12, and why $\tilde{C}$ deserves the name “distortion”). Figure 2.12 shows two material curves meeting at $\tilde{X}$ at angle Θ in κ , and their images meeting at $\tilde{x}$ at angle θ in κ_t . Formulas (2.106) and (2.108) say that $\tilde{C}$ (or $\tilde{B}$) determines *every* stretch and *every* angle change at the point, and these are the entire local geometric content of

a deformation: hence the Cauchy–Green tensors are said to characterise the *state of distortion* at p . Note the shear reading of the off-diagonal entries: taking $\tilde{M}_1 = \hat{\tilde{E}}_1$, $\tilde{M}_2 = \hat{\tilde{E}}_2$ (initially orthogonal, $\Theta = \pi/2$), (2.106) collapses to $\cos \theta = C_{12}/\sqrt{C_{11}C_{22}}$, so a nonzero C_{12} is the loss of perpendicularity, while the diagonal entries $C_{AA} = \mu^2$ (no sum) are squared stretches along the axes.

2.8 Problems

The three problems of §2.8 are solved below in full index notation, using only Chapter 1 algebra together with (2.39), $F_{iA} = \chi_{i,A}$. Recall the referential and spatial permutation symbols $\varepsilon_{ABC} = [\hat{\tilde{E}}_A, \hat{\tilde{E}}_B, \hat{\tilde{E}}_C]$ and $\varepsilon_{ijk} = [\underline{e}_i, \underline{e}_j, \underline{e}_k]$, and the contracted identities $\varepsilon_{ABC}\varepsilon_{DBC} = 2\delta_{AD}$, $\varepsilon_{ijk}\varepsilon_{ijk} = 6$.

LEMMA 2.36 (symmetric contracted with antisymmetric vanishes). *If $S_{BC} = S_{CB}$ and $A_{BC} = -A_{CB}$, then $A_{BC}S_{BC} = 0$, the sum running over both indices.*

Proof. Use the antisymmetry of A , then relabel the two dummy indices, then use the symmetry of S :

$$A_{BC}S_{BC} \stackrel{(1)}{=} -A_{CB}S_{BC} \stackrel{(2)}{=} -A_{BC}S_{CB} \stackrel{(3)}{=} -A_{BC}S_{BC}.$$

Step (1) is $A_{BC} = -A_{CB}$; step (2) renames the summation indices; step (3) is $S_{CB} = S_{BC}$. Hence $2A_{BC}S_{BC} = 0$. ■

The lemma is used twice below, at 2(a) and at 2(b).

Problem 1: components of the cofactor of $\tilde{F}$. *Show that $F_{iA}^* = \frac{1}{2}\varepsilon_{ijk}\varepsilon_{ABC}F_{jB}F_{kC}$.*

Solution. The cofactor is defined by $\tilde{F}^*(\underline{a} \times \underline{b}) = \tilde{F}\underline{a} \times \tilde{F}\underline{b}$ for all $\underline{a}, \underline{b} \in \hat{\mathbb{E}}^3$. Take $\underline{a} = \hat{\tilde{E}}_B$, $\underline{b} = \hat{\tilde{E}}_C$, with

$$\hat{\tilde{E}}_B \times \hat{\tilde{E}}_C = \varepsilon_{BCA}\hat{\tilde{E}}_A$$

by Proposition 1.24. The left side becomes

$$\tilde{F}^*(\varepsilon_{BCA}\hat{\tilde{E}}_A) = \varepsilon_{BCA}\tilde{F}^*\hat{\tilde{E}}_A = \varepsilon_{BCA}F_{iA}^*\varepsilon_i,$$

using $\tilde{F}^*\hat{\tilde{E}}_A = F_{iA}^*\varepsilon_i$ (the analogue of (2.40)). The right side, expanding each image on $\{\varepsilon_i\}$ and using $\varepsilon_j \times \varepsilon_k = \varepsilon_{jki}\varepsilon_i$, is

$$\tilde{F}\hat{\tilde{E}}_B \times \tilde{F}\hat{\tilde{E}}_C = (F_{jB}\varepsilon_j) \times (F_{kC}\varepsilon_k) = F_{jB}F_{kC}\varepsilon_{jki}\varepsilon_i.$$

Equating components on ε_i and writing $\varepsilon_{jki} = \varepsilon_{ijk}$ (cyclic),

$$\varepsilon_{BCA}F_{iA}^* = \varepsilon_{ijk}F_{jB}F_{kC}.$$

Contract both sides with ε_{BCD} (sum over B, C). On the left, $\varepsilon_{BCA}\varepsilon_{BCD} = 2\delta_{AD}$, giving $2\delta_{AD}F_{iA}^* = 2F_{iD}^*$; on the right, $\varepsilon_{BCD} = \varepsilon_{DBC}$ (cyclic). Hence

$$F_{iD}^* = \frac{1}{2}\varepsilon_{ijk}\varepsilon_{DBC}F_{jB}F_{kC},$$

which upon renaming $D \rightarrow A$ is the required $F_{iA}^* = \frac{1}{2}\varepsilon_{ijk}\varepsilon_{ABC}F_{jB}F_{kC}$.

□

The identity $\varepsilon_{BCA}\varepsilon_{BCD} = 2\delta_{AD}$ is Theorem 1.29. Directly: for $A \neq D$ every term repeats an index and dies; for $A = D = 3$ the survivors are $\varepsilon_{123}^2 + \varepsilon_{213}^2 = 2$. The factor 2 is why $\frac{1}{2}$ appears, each unordered pair $\{B, C\}$ being counted twice.

Check. For the simple shear of Chapter 3, $F_{iA} = \delta_{iA} + \gamma\delta_{i1}\delta_{A2}$, whose matrix is $\begin{pmatrix} 1 & \gamma & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. Since $\det \tilde{F} = 1$ and $\tilde{F}^* = J\tilde{F}^{-t} = \tilde{F}^{-t}$ by (2.55), the answer must be $\begin{pmatrix} 1 & 0 & 0 \\ -\gamma & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. Test the formula on the entry $A = 1, i = 2$:

$$F_{21}^* = \frac{1}{2}\varepsilon_{2jk}\varepsilon_{1BC}F_{jB}F_{kC}.$$

The only non-zero ε_{2jk} are $\varepsilon_{231} = 1$ and $\varepsilon_{213} = -1$; the only non-zero ε_{1BC} are $\varepsilon_{123} = 1$ and $\varepsilon_{132} = -1$. So four terms survive:

$$\begin{aligned} 2F_{21}^* &= \varepsilon_{231}\varepsilon_{123}F_{32}F_{13} + \varepsilon_{231}\varepsilon_{132}F_{33}F_{12} \\ &\quad + \varepsilon_{213}\varepsilon_{123}F_{12}F_{33} + \varepsilon_{213}\varepsilon_{132}F_{13}F_{32} \\ &= (+1)(+1)(0)(0) + (+1)(-1)(1)(\gamma) \\ &\quad + (-1)(+1)(\gamma)(1) + (-1)(-1)(0)(0) \\ &= -\gamma - \gamma = -2\gamma, \end{aligned}$$

so $F_{21}^* = -\gamma$, as required.

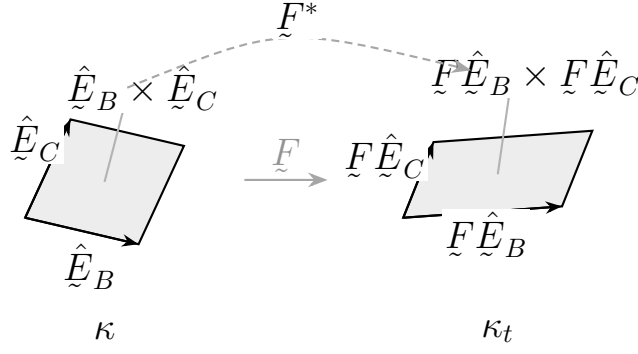


Figure 2.13: Problem 1. $\tilde{F}$ carries the edge vectors of a material parallelogram, $\tilde{F}^*$ carries its area vector. The problem converts the implicit definition into components, $F_{iA}^* = \frac{1}{2}\varepsilon_{ijk}\varepsilon_{ABC}F_{jB}F_{kC}$; (2.55) then converts those into $J\tilde{F}^{-t}$.

Problem 2(a): F_{iA}^* as a referential divergence. Show $F_{iA}^* = G_{iAB,B}$, where $G_{iAB} = \frac{1}{2}\varepsilon_{ijk}\varepsilon_{ABC}\chi_j\chi_{k,C}$.

Solution. Differentiate G_{iAB} with respect to X_B and sum, using the product rule (the ε 's are constants):

$$G_{iAB,B} = \frac{1}{2}\varepsilon_{ijk}\varepsilon_{ABC}(\chi_{j,B}\chi_{k,C} + \chi_j\chi_{k,CB}).$$

Second term: for fixed A, k put $S_{BC} = \chi_{k,CB}$ and $A_{BC} = \varepsilon_{ABC}$. Mixed partials commute, so $S_{BC} = S_{CB}$; ε changes sign under a swap, so $A_{BC} = -A_{CB}$. By Lemma 2.36,

$$\varepsilon_{ABC}\chi_{k,CB} = 0.$$

First term, by $F_{jB} = \chi_{j,B}$ and Problem 1,

$$\frac{1}{2}\varepsilon_{ijk}\varepsilon_{ABC}F_{jB}F_{kC} = F_{iA}^*.$$

Hence $G_{iAB,B} = F_{iA}^*$. □

Problem 2(b): the Piola identity. Show $G_{iAB} = -G_{iBA}$ and deduce $F_{iA,A}^* = 0$.

Solution. Swapping the free indices A, B in the definition and using only the antisymmetry $\varepsilon_{BAC} = -\varepsilon_{ABC}$,

$$G_{iBA} = \frac{1}{2}\varepsilon_{ijk}\varepsilon_{BAC}\chi_j\chi_{k,C} = -\frac{1}{2}\varepsilon_{ijk}\varepsilon_{ABC}\chi_j\chi_{k,C} = -G_{iAB}.$$

Now take the referential divergence of Problem 2(a):

$$F_{iA,A}^* = (G_{iAB,B})_{,A} = G_{iAB,BA}.$$

Apply Lemma 2.36 again on the pair (A, B) : $(\cdot)_{,BA}$ is symmetric, G_{iAB} antisymmetric. Hence

$$F_{iA,A}^* = 0, \quad \text{i.e.} \quad \text{Div } \tilde{F}^* = \mathfrak{O},$$

the *Piola identity*. □

$$\int_{\partial P} \tilde{F}^* \tilde{N} \, dA = \int_P \text{Div } \tilde{F}^* \, dV = \mathfrak{O} \quad \Longleftrightarrow \quad \int_{\partial P_t} \tilde{n} \, da = \mathfrak{O}$$

Figure 2.14: The Piola identity. A closed material surface stays closed, so its normals integrate to zero in κ_t as they do in κ . By Piola–Nanson (2.54) that is the divergence theorem applied to $\tilde{F}^*$, and it holds for every P and every motion only if $\text{Div } \tilde{F}^* = \mathfrak{O}$.

REMARK 2.37 (what the Piola identity buys). $\text{Div } \tilde{F}^* = \mathfrak{O}$ says the co-factor field is automatically divergence free, no matter what motion is prescribed. Combined with Piola–Nanson (2.54), it is exactly the statement that the deformed images of closed material surfaces stay closed: applying the divergence theorem, $\int_{\partial P} \tilde{F}^* \tilde{N} \, dA = \int_P \text{Div } \tilde{F}^* \, dV = \mathfrak{O}$, i.e. $\int_{\partial P_t} \tilde{n} \, da = \mathfrak{O}$, the elementary fact that the outward normals of any closed surface integrate to zero. It is the kinematic identity that later converts the Cauchy stress balance into its Piola form.

Problem 2(c): $\det \tilde{F}$ as a divergence. Show $\det \tilde{F} = \frac{1}{3} F_{iA} F_{iA}^*$, and hence that there is a field $H(X, t) = H_A \hat{E}_A$ with $\det \tilde{F} = H_{A,A}$.

Solution. The cofactor identity $\underline{A}^t \underline{A}^* = (\det \underline{A}) \underline{I}$ of Problem 22(c) of Chapter 1 holds verbatim for the two-point $\underline{F}$, where it reads, in components,

$$F_{iA} F_{iB}^* = (\det \underline{F}) \delta_{AB}.$$

Setting $B = A$ and summing, with $\delta_{AA} = 3$,

$$F_{iA} F_{iA}^* = 3 \det \underline{F} \quad \implies \quad \det \underline{F} = \frac{1}{3} F_{iA} F_{iA}^*.$$

Now substitute $F_{iA} = \chi_{i,A}$ and use the Piola identity of part (b) to move the derivative outside:

$$\det \underline{F} = \frac{1}{3} \chi_{i,A} F_{iA}^* = \frac{1}{3} \left[(\chi_i F_{iA}^*)_{,A} - \chi_i \underbrace{F_{iA,A}^*}_{=0} \right] = \left(\frac{1}{3} \chi_i F_{iA}^* \right)_{,A}.$$

Hence $\det \underline{F} = H_{A,A}$ with

$$H_A = \frac{1}{3} \chi_i F_{iA}^*,$$

i.e. $\underline{H} = \frac{1}{3} \underline{F}^{*t} \underline{\chi}_\kappa$. □

REMARK 2.38 (why one wants J as a divergence). Writing $J = \det \underline{F} = \text{Div } \underline{H}$ turns the volume integral of J into a surface integral by the divergence theorem, $\int_P J \, dV = \int_{\partial P} \underline{H} \cdot \underline{N} \, dA$. Since $\int_P J \, dV = \int_{P_t} dv$ is the deformed volume, this expresses the volume enclosed by a deformed surface purely in terms of data *on* that surface, the familiar “volume from a surface integral” trick, and it is the kinematic origin of several variational identities in nonlinear elasticity.

Problem 2(d): $\det \underline{F}^t = \det \underline{F}$. *Solution.* Substituting Problem 1 into the result of 2(c),

$$\det \underline{F} = \frac{1}{3} F_{iA} \cdot \frac{1}{2} \varepsilon_{ijk} \varepsilon_{ABC} F_{jB} F_{kC} = \frac{1}{6} \varepsilon_{ijk} \varepsilon_{ABC} F_{iA} F_{jB} F_{kC}, \quad (*)$$

the two-point version of the contracted determinant formula of Chapter 1. The transpose has components $(\underline{F}^t)_{Ai} = F_{iA}$, and $\underline{F}^t$ maps

$\mathbb{E}^3 \rightarrow \hat{\mathbb{E}}^3$, so the same formula with the roles of the two index families exchanged gives

$$\begin{aligned}\det \underline{F}^t &= \frac{1}{6} \varepsilon_{ABC} \varepsilon_{ijk} (\underline{F}^t)_{Ai} (\underline{F}^t)_{Bj} (\underline{F}^t)_{Ck} \\ &= \frac{1}{6} \varepsilon_{ABC} \varepsilon_{ijk} F_{iA} F_{jB} F_{kC} = \det \underline{F},\end{aligned}$$

since the two scalar expressions differ only in the order in which the same factors are written. $\square$

Problem 3: change of basis for a two-point tensor. *With $\{\hat{E}_A\}, \{\hat{E}'_A\}$ orthonormal in $\hat{\mathbb{E}}^3$ and $\{\underline{e}_i\}, \{\underline{e}'_i\}$ orthonormal in $\mathbb{E}^3$, and $\underline{F} = F_{iA} \underline{e}_i \otimes \hat{E}_A = F'_{iA} \underline{e}'_i \otimes \hat{E}'_A$, find F'_{iA} .*

Solution. Introduce the two independent sets of direction cosines, one for each space,

$$a_{ij} = \underline{e}_i \cdot \underline{e}'_j \quad (\text{spatial}), \quad \hat{a}_{AB} = \hat{E}_A \cdot \hat{E}'_B \quad (\text{referential}),$$

so that, by the orthonormal expansion of Chapter 1,

$$\underline{e}'_i = (\underline{e}_j \cdot \underline{e}'_i) \underline{e}_j = a_{ji} \underline{e}_j, \quad \hat{E}'_A = (\hat{E}_B \cdot \hat{E}'_A) \hat{E}_B = \hat{a}_{BA} \hat{E}_B.$$

Extract the primed components with (2.41), $F'_{iA} = \underline{e}'_i \cdot \underline{F} \hat{E}'_A$. First act with $\underline{F}$ on the primed referential basis vector, using linearity and $\underline{F} \hat{E}_B = F_{jB} \underline{e}_j$ from (2.40):

$$\underline{F} \hat{E}'_A = \hat{a}_{BA} \underline{F} \hat{E}_B = \hat{a}_{BA} F_{jB} \underline{e}_j.$$

Then dot with $\underline{e}'_i = a_{ki} \underline{e}_k$ and use $\underline{e}_k \cdot \underline{e}_j = \delta_{kj}$:

$$F'_{iA} = a_{ki} \underline{e}_k \cdot \hat{a}_{BA} F_{jB} \underline{e}_j = a_{ki} \hat{a}_{BA} F_{jB} \delta_{kj} = a_{ji} \hat{a}_{BA} F_{jB}.$$

Hence

$$\boxed{F'_{iA} = a_{ji} \hat{a}_{BA} F_{jB}}$$

$\square$

In matrix form. With $(a)_{ij} = a_{ij}$ and $(\hat{a})_{AB} = \hat{a}_{AB}$, the index pattern $a_{ji}(\cdot)_{jB} \hat{a}_{BA}$ is

$$(F') = (a)^t (F) (\hat{a}).$$

Both (a) and $(\hat{a})$ are orthogonal, by the computation of Problem 9(a) of Chapter 1 applied in each space separately:

$$a_{ki}a_{kj} = \delta_{ij}, \quad \hat{a}_{CA}\hat{a}_{CB} = \delta_{AB}.$$

Check 1: J is unchanged. Taking determinants,

$$\det(F') = \det(a)^t \det(F) \det(\hat{a}) = (\pm 1) \det(F) (\pm 1),$$

and both signs are $+1$ when the two changes of basis preserve handedness. So J is basis independent, as (2.50) requires.

Check 2: $\tilde{C}$ transforms with one matrix only. From $C_{AB} = F_{iA}F_{iB}$,

$$\begin{aligned} C'_{AB} &= F'_{iA}F'_{iB} = (a_{ji}\hat{a}_{CA}F_{jC})(a_{ki}\hat{a}_{DB}F_{kD}) \\ &= \hat{a}_{CA}\hat{a}_{DB}\delta_{jk}F_{jC}F_{kD} = \hat{a}_{CA}\hat{a}_{DB}C_{CD}, \end{aligned}$$

using $a_{ji}a_{ki} = \delta_{jk}$. The spatial cosines have cancelled: $\tilde{C}$ is a referential tensor and cannot feel a change of spatial basis. The same computation on $B_{ij} = F_{iA}F_{jA}$ cancels the referential cosines instead.

REMARK 2.39 (the transformation law identifies the tensor's type). Compare with Chapter 1: a vector picks up one factor of the cosine matrix, $v'_i = a_{ki}v_k$; an ordinary (one-point) tensor picks up two factors of the *same* matrix, $A'_{ij} = a_{ki}a_{lj}A_{kl}$. A two-point tensor also picks up two factors, but one from *each* space, a_{ji} for the lower-case index and $\hat{a}_{BA}$ for the upper-case one. The two rotations are independent: one may re-label the reference frame without touching the current frame, and $\tilde{F}$ responds only in its capital index. This is the precise sense in which $\tilde{F}$ “has one foot in each configuration”, and it is why the alphabet convention of Remark 2.10 is more than cosmetic.

Appendix A

Dictionary to the book

Book	Here	Book	Here
(2.1) motion	Def. 2.2	(2.34)–(2.37)	§2.4, Thm. 2.14
(2.2) velocity, accel.	Def. 2.3	(2.38)–(2.41)	eqs. here in §2.4
(2.3), (2.4) reference	Def. 2.4	(2.42)–(2.44)	Lem. 2.15
(2.5)–(2.7) χ_κ	eqs. in §2.1	(2.45), (2.46)	material curves, §2.4
(2.8)–(2.10) velocity forms	§2.1	(2.47)–(2.50) volume	eqs. here, Rem. 2.18
(2.11)–(2.13) descriptions	§2.2, Rem. 2.6	(2.51)–(2.54) Piola–Nanson	§2.4
(2.14)–(2.16) differentials	§2.2	(2.55) $F^* = JF^{-t}$	§2.4
(2.17)–(2.21) material deriv.	Thm. 2.8	(2.56)–(2.59) mass	§2.5
(2.22)–(2.24) coordinates	§2.3, Rem. 2.10	(2.60)–(2.65) localization	Thm. 2.20
(2.25)–(2.29) components	§2.3	Example (2.30)–(2.33)	Example 2.12
(2.66), (2.67) stretch	§2.6, Rem. 2.22	(2.89)–(2.95) $\underline{B}$	Prop. 2.28
(2.68)–(2.74) shifter	Def. 2.23, Prop. 2.24	(2.96)–(2.99) strains	Def. 2.31, Rem. 2.32
(2.75)–(2.78) displacement	§2.6, Rem. 2.25	(2.100)–(2.103) arclength	§2.7, Rem. 2.34
(2.79)–(2.81) undeformed	Prop. 2.26	(2.104)–(2.108) angles	§2.7, Rem. 2.35
(2.82)–(2.88) $\underline{C}$	Prop. 2.27	§2.8 Problems 1–3	§2.8

Figures 2.1–2.12 are described in Remarks 2.1, 2.5, 2.7, 2.10, 2.16, 2.18, 2.19, 2.22, 2.34 and 2.35. All algebra uses only Chapter 1: the tensor product and its component rule, the dyad product $(\underline{a} \otimes \underline{b})(\underline{c} \otimes \underline{d}) = (\underline{b} \cdot \underline{c})\underline{a} \otimes \underline{d}$, the transpose rules of Problems 17–20, the box product and determinant $[\underline{F}\underline{a}, \underline{F}\underline{b}, \underline{F}\underline{c}] = \det \underline{F} [\underline{a}, \underline{b}, \underline{c}]$, the cofactor identity $\underline{A}^t \underline{A}^* = (\det \underline{A})\underline{I}$ of Problem 22(c), the contracted ε – δ identities, and the coordinate free chain rule $d f = \text{grad } f \cdot d\mathbf{x}$.